

Spectral induced polarization of shaly sands: Influence of the electrical double layer

A. Revil^{1,2}

Received 9 August 2011; revised 4 January 2012; accepted 5 January 2012; published 16 February 2012.

[1] I developed a new model named POLARIS describing the complex conductivity of (pyrite-free) shaly poorly sorted sands. This model is based on the solution given by the effective medium theory for grains coated by an electrical double layer and immersed in a background electrolyte. The electrical double layer comprises the Stern layer and the diffuse layer. Both layers play very distinct roles in the in-phase and quadrature conductivities. The polarization of the shaly sands is mainly controlled by the polarization of the Stern layer (except at very high salinities) with a very small mobility of the counterions contained in this layer. The in-phase component of the conductivity is controlled by the conductivity of the pore water with a contribution associated with the diffuse layer (the contribution of the Stern layer seems negligible). The fraction of counterions in the Stern layer is computed from a simple sorption isotherm and is used to infer the quadrature conductivity. The quadrature conductivity is assumed to be frequency independent, which is a reasonable approximation in clayey sands and sandstones, in agreement with observations. The polarization model is also based on the assumption that the Stern layer is discontinuous between grains, an assumption that is consistent with recent models of ionic transport in clayey sands. POLARIS explains the dependence of the quadrature conductivity on the salinity, cation exchange capacity, specific surface area (or specific surface per unit pore volume), and temperature. It can be used to predict the saturation and the permeability (inside 1 order of magnitude).

Citation: Revil, A. (2012), Spectral induced polarization of shaly sands: Influence of the electrical double layer, *Water Resour. Res.*, 48, W02517, doi:10.1029/2011WR011260.

1. Introduction

[2] Induced Polarization is a geophysical method that is sensitive to the ability of porous materials to store reversibly electrical charges. In environmental geophysics, induced polarization offers an extension to the classical direct current (DC) conductivity/resistivity method to image soils and shallow aquifers [Vanhal, 1997; Morgan *et al.*, 1999; Binley *et al.*, 2005]. In frequency domain induced polarization, in addition to a conductivity magnitude, a phase lag is measured between the current and the electrical field. This phase lag represents a polarization over a conduction term (like the chargeability in time domain induced polarization). The conductivity magnitude and the phase lag can be used to define a complex conductivity in which both an in-phase conductivity and a quadrature conductivity contribution can be defined [e.g., Olhoeft, 1986]. The complex conductivity can be investigated over a broad range of frequencies (typically from 1 mHz to 10 kHz).

[3] For clayey sands, the conductivity and the phase lag are observed to be quite frequency independent over a

broad range of frequencies (e.g., in the range 10 mHz to 100 Hz in the work by Vinegar and Waxman [1984] (their Figures 4 and 5 for clayey sands and sandstones) and Slater and Glaser [2003] (their Figure 3)). A constant phase lag behavior has been usually connected to the fractal nature of the material (see Hyslip and Vallejo [1997] and Drolon *et al.* [2003] for the fractal characterization of sand grains and Le Méhauté and Crépy [1983] and Wong [1987] for fractal-based low-frequency polarization models). A broad range of polarization length scales covering several decades can be responsible for this behavior. In some cases, however, shaly sands do show distinct frequency dependence with a distribution of relaxations around some central frequency. The associated relaxation times may be determined by the pore throat size of the material [Scott and Barker, 2003; Binley *et al.*, 2005; Zisser *et al.*, 2010] or possibly by the grain size distribution [e.g., Revil and Florsch, 2010].

[4] In the present paper, I will focus on the first class of materials exhibiting mainly a frequency-independent polarization (over the frequency range 10 mHz to 100 Hz). The case of materials exhibiting dispersion peak(s) will be examined in a future work. In addition, I am dealing with sandstones free of non-oxidized pyrite [see Pelton *et al.*, 1978]. The reason is that pyrite is responsible for a very strong polarization depending on its concentration and grain size. Because some authors do not report the pyrite content of their samples, it is possible that some observations made

¹Department of Geophysics, Colorado School of Mines, Golden, Colorado, USA.

²ISTerre, UMR CNRS 5275, Université de Savoie, Le Bourget du Lac, France.

in the literature of well-characterized phase peak could be due to the presence of pyrite.

[5] For a long time, the development of the induced polarization method has been slowed down by the lack of physical understanding of the measured response. In geophysics, the induced polarization response is usually modeled with empirical responses (like the Cole-Cole or the constant phase angle models). The use of these empirical models is limited because they have no predictive powers. In other words, they are merely a parametric way to represent the complex conductivity spectra using only few parameters.

[6] Several mechanistic models have been proposed in the literature but none of them have been able to explain, in a consistent way, all the observations available to date. These mechanistic models are usually divided into two groups: (1) polarization models of the electrical double layer [Schwarz, 1962; de Lima and Sharma, 1992; Lesmes and Morgan, 2001; Leroy et al., 2008] and (2) the membrane polarization model [Marshall and Madden, 1959; Dukhin and Shilov, 2002; Titov et al., 2002]. This separation is in reality very arbitrary as all the double layer polarization models developed in colloidal chemistry include a membrane polarization component, which involves the formation of neutral clouds of ions on both the side of a particle or the side of a pore throat in presence of a harmonic electrical field.

[7] In the recent literature, Revil and coworkers [Leroy et al., 2008; Revil and Florsch, 2010; Jougnot et al., 2010; Schmutz et al., 2010; Vaudelet et al., 2011a, 2011b; Revil and Skold, 2011] have developed a Stern polarization model, which has been used successfully to explain a number of observations made in the laboratory, especially for silica sands. However, as explained by Revil and Florsch [2010], there are some strong differences between the electrical double layer associated with silica grains and clay minerals. In the case of silica grains, the fraction of counterions populating the Stern layer is very salinity-dependent and usually smaller than 60%. Therefore, there is a lot of room for the counterions to move along the mineral surface in this layer. It was found that the mobility of the counterions in the Stern layer of silica grains is quite similar (if not equal) to the mobility of the same ions in the free water [Tarasov and Titov, 2007; Revil and Florsch, 2010].

[8] In clay minerals, the fraction of counterions in the Stern layer is very high (between 80% and 98% [Leroy and Revil, 2009, and references therein]). I expect therefore that the mobility of the counterions in the Stern layer of clays may be much smaller than in the bulk pore water because of the high density of counterions populating the Stern layer. What is the impact of this low mobility on the overall complex conductivity of shaly sands? One may think that this question could be answered independently by molecular dynamics (see, for instance, Bourg and Sposito [2011], who found that the mobility of the counterions in the Stern layer is roughly a 10th of the mobility in the pore water). That said, computing the mobility of the counterions in the Stern layer from molecular dynamics is subject to many assumptions and therefore this approach does not yield, yet, compelling values, as explained by Bourg and Sposito [2011].

[9] I developed below a new model named POLARIS on the basis of the effective medium theory. This type of upscaling method has been very successful to model the

conductivity of shaly sands [Bussian, 1983; de Lima and Sharma, 1990; Revil et al., 1998]. I will explain why the surface conductivity of the clay particles is dominated by the diffuse layer and the polarization of the clay particles by the Stern layer. POLARIS seems to explain an unprecedented number of observations with a minimum of input parameters. The end goal of this model is to interface between reactive transport modeling and geophysical time domain or frequency domain induced polarization. Also my goal in developing POLARIS is to keep this model simple enough so it can be used with a minimum of assumptions.

2. Complex Conductivity

[10] In section 2 I will define first the conventions used below to describe the electrical conductivity. Then, I will describe the result of the differential effective medium theory (DEM) for the in-phase conductivity (section 2.1) and its implication in terms of quadrature conductivity (section 2.2). I will show that the in-phase conductivity has a surface conductivity term dominated by the diffuse layer while the quadrature conductivity is dominated by the Stern layer. Therefore the determination of the partition coefficient of the counterions between the diffuse and the Stern layer is of paramount importance in POLARIS. I will develop in section 2.3 a simple sorption isotherm to compute this partition coefficient as a function of the pH (for near neutral pH values) and salinity.

[11] I start by defining the convention I use below. Ampère's law is written as

$$\nabla \cdot \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}, \quad (1)$$

where t is time (in s), $\mathbf{J}$ is the conduction (electromigration) current density, $\mathbf{J}_d = \partial \mathbf{D} / \partial t$ denotes the displacement current density (both in A m^{-2} , the subscript d is for displacement), $\mathbf{H}$ is the magnetic field (expressed in A m^{-1}), $\mathbf{D} = \epsilon^* \mathbf{E}$ is the dielectric displacement (in C m^{-2}), and ϵ^* is the permittivity (in F m^{-1}) of the material (ϵ^* / ϵ_0 denotes the relative complex permittivity and ϵ_0 the permittivity of vacuum). In linear models, the current density is given by a linear Ohm's law, $\mathbf{J} = \sigma^* \mathbf{E}$ with the harmonic electrical field $\mathbf{E} = \mathbf{E}_0 \exp(i\omega t)$, $i = \sqrt{-1}$ the pure imaginary number, and ω is the angular frequency with $\mathbf{E}_0$ a constant electrical field magnitude and direction. As a side note, the linear Ohm's law (i.e., a linear constitutive relationship between the current density and the electrical field) does not hold in presence of a membrane polarization mechanism except for small electrical fields [see, e.g., Urtenov et al., 2007; Sistat et al., 2008, and reference therein]. The total (subscript t) current density is the sum of the conduction current and displacement current densities. This yields $\mathbf{J}_t = (\sigma^* + i\omega\epsilon^*)\mathbf{E}$ [Vinegar and Waxman, 1984] where σ^* and ϵ^* are complex scalars dependent upon frequency, $\sigma^* = \sigma' + i\sigma''$ and $\epsilon^* = \epsilon' + i\epsilon''$. The total current density can be written as

$$\mathbf{J}_t = \sigma_{\text{eff}}^* \mathbf{E}, \quad (2)$$

where $\sigma_{\text{eff}}^* = \sigma_{\text{eff}} + i\omega\epsilon_{\text{eff}}$ is the effective or apparent complex conductivity and σ_{eff} and ϵ_{eff} are real scalars dependent upon frequency. These effective parameters are the

parameters that are measured during an experiment in the laboratory or in the field but these terms contain both electromigration and true dielectric polarization effects. They are given by

$$\sigma_{eff} = \sigma' - \omega\epsilon'' \quad (3)$$

$$\epsilon_{eff} = \epsilon' + \sigma''/\omega \quad (4)$$

(note that $-\omega\epsilon''$ is always positive). Therefore, the effective conductivity and permittivity include contributions from charge conduction and polarization mechanisms. In this paper, I am only interested in the complex conductivity,

$$\sigma^* = \sigma' + i\sigma'', \quad (5)$$

which dominates the measured response at low frequencies (below 100 Hz). With the sign conventions taken above, the quadrature conductivity is negative as well as the phase lag (other conventions exist). In the following, I will consider that the in-phase conductivity and the quadrature conductivity are frequency independent. Actually, their frequency dependence is very small and can be modeled with the constant phase lag model, as shown by Börner [1992].

[12] I denote $\omega = 2\pi f$ the angular frequency, $f = \omega/(2\pi)$ the frequency (in Hertz). The magnitude of the conductivity $|\sigma^*|$ and the phase lag φ are related to the real (in-phase) and imaginary (out-of-phase or quadrature) components of the complex conductivity σ^* , σ' , and σ'' (expressed in $S\ m^{-1}$) by

$$\sigma^* = \sigma' + i\sigma'' = |\sigma| \exp(i\varphi), \quad (6)$$

with $|\sigma^*| = (\sigma'^2 + \sigma''^2)^{1/2}$ and $\tan \varphi = \sigma''/\sigma'$. Note that some researchers use an alternative convention for the complex conductivity $\sigma^* = \sigma' - i\sigma''$, which yields a positively-defined quadrature conductivity [Olhoeft, 1979; Börner, 1992; Kemna, 2000].

[13] I introduce now the physical concepts used to describe the induced polarization model developed below. The first one is that the clay particles are coated by an electrical double layer model. These two layers include the Stern layer of sorbed counterions and the diffuse layer (Figure 1). This double layer model of clay particles is rather well established [see Leroy and Revil, 2004, and references therein] and the existence of the Stern layer is in agreement with molecular dynamic simulations showing a condensation of the counterions directly on the mineral surface [Wang et al., 2007; Tournassat et al., 2009]. We consider two types of granular porous materials: Either the porous material itself is made of grains of different sizes with the surface area dominated by the presence of clay particles (type I) or the material is made of silica grains, no clays, and the silica grains are assumed to be relatively well sorted (type II) (Figure 2a). I consider in both cases that the porous material is isotropic and homogeneous at the scale of a representative elementary volume (REV) and fully saturated by a pore water solution with hydrated ions. The mineral grains have some contiguity, which prevents a strong polarization of the diffuse layer. That said, the Stern layer remains discontinuous and therefore the distribution of grain size can represent a distribution of polarization length scales. In addition, for such dense grain assemblage,

the grain-to-grain interactions cannot be limited to a single dipole term. As a result, the distribution of relaxation times is expected to be very broad. The importance of the Stern layer in the polarization processes of porous media has been underlined by many authors [see Revil and Florsch, 2010; Grosse, 2011, and references therein]. The polarization of this layer (called also the stagnant layer in electrochemistry) is central to my model. The broad distribution of the polarization length scales implies weakly frequency-dependent phase lag and quadrature conductivity for type I materials (Figure 2b). At the opposite, for type II materials, a clear phase peak can be observed (Figure 2c). In the following, I am only interested in phase I materials.

[14] The second point is related to the upscaling approach used to obtain the macroscopic electrical conductivity at the scale of the REV. For the type of porous material I am dealing with (granular with a broad distribution of grain sizes; see Figure 2a), the solution of the DEM is effective for describing the macroscopic conductivity [e.g., Sen et al., 1981; Bussian, 1983; de Lima and Sharma, 1990, 1992; Revil et al., 1998]. Using the DEM, the complex conductivity can be described as a function of the conductivity of the pore water σ_f and the (complex) surface conductivity σ_s^* by [e.g., Sen et al., 1981]

$$\sigma^* = \frac{\sigma_f}{F} \left(\frac{1 - \sigma_s^*/\sigma_f}{1 - \sigma_s^*/\sigma^*} \right)^m, \quad (7)$$

$$F = \phi^{-m}, \quad (8)$$

where m is called the cementation or Archie exponent (dimensionless) [Archie, 1942], F is the electrical formation factor (dimensionless), ϕ is the connected porosity (dimensionless), σ_f (in $S\ m^{-1}$) denotes the conductivity of the pore water (real number), and $\sigma_s^*(\omega)$ denotes the (complex-valued) surface conductivity. Equation (7) is also known as the Bruggeman-Hanai equation [Bruggeman, 1935; Hanai, 1968]. Equation (8) is generally called Archie's law in the literature [Archie, 1942]. This deserves here three short comments: (1) First, there is no prefactor a in Archie's law as defined by Archie himself in 1942. (2) Archie's law *stricto sensu* is dealing with the linear correlation in a log-log plot between the formation factors and porosities for a collection of samples extracted from the same formation; this is not the case in equation (8) in which m is defined for a single material. (3) Equation (8) connects the inverse of the formation factor (which is exactly an effective porosity [see Revil and Cathles, 1999]) to the porosity considered to be a geometrical parameter. Therefore, m is perceived physically as a decoupling parameter between the total connected porosity and the fraction of this porosity dynamically connected by the electrical field lines.

[15] For shaly sands, the value of the cementation exponent can be predicted from the clay mineralogy and the clay content as discussed, for instance, by Revil et al. [1998, Figure 5]. The choice of the DEM is justified by the statistical analysis performed by Niwas et al. [2007] showing this model is much superior to the "generalized" Archie's law developed by Glover et al. [2000]. Also, as discussed by many authors [e.g., Sen and Goode, 1992, and references therein], the Waxman and Smits [1968] equation has an

Electrical double layer of clay minerals

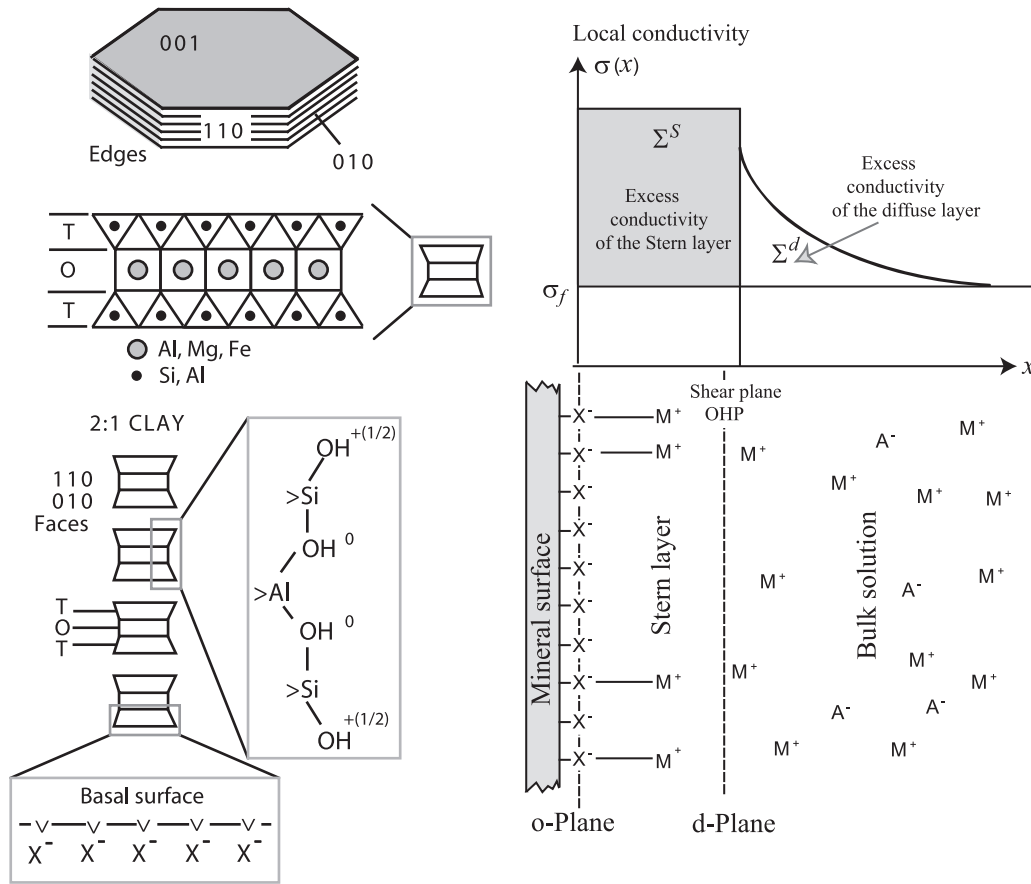


Figure 1. Electrical double layer at the surface of clays. The surface charge of the clays is heterogeneous and results from amphoteric sites located on the 110 and 010 crystalline planes, while the basal surfaces 001 carry negative sites associated with isomorphic substitutions in the crystalline framework (especially for illite and smectite). The electrical double layer is formed by the Stern layer of sorbed counterions (on the 110 and 010 planes) and the diffuse layer. The o plane defines the mineral surface, while the d plane defines the interface between the Stern layer and the diffuse layer on the 110 and 010 crystalline planes. The specific surface conductivities Σ_S and Σ_d define the excess surface conductivity associated with the Stern and diffuse layers with respect to the conductivity of the pore water.

unphysical dependence of the apparent mobility of the counterions with the salinity. *Revil et al.* [1998] performed a statistical analysis of the DEM versus the Waxman and Smits model and found that the DEM is also a better model to fit laboratory data avoiding such unphysical dependence for the mobility of the counterions.

2.1. In-Phase Conductivity

[16] *Revil et al.* [1998] used the DEM approach to find the in-phase component of the complex conductivity using an approximate analytical solution of the DEM at the scale of a REV. *Revil et al.* [2002] modified slightly this equation to account for the diffuse layer nature of surface conductivity at very low salinities. Their model yields the following expressions for the in-phase conductivity:

$$\sigma' = \sigma_f Du, Du \geq 1, \quad (9)$$

$$\sigma' = \frac{\sigma_f}{F} \left[F Du + \frac{1}{2} (1 - Du) \left(1 - Du + \sqrt{(1 - Du)^2 + 4 F Du} \right) \right], \quad Du \leq 1, \quad (10)$$

where Du is a dimensionless number called the Dukhin number [e.g., *Bolève et al.*, 2007]. This Dukhin number represents the ratio between the in-phase surface conductivity of the grains to the electrical conductivity of the pore water [e.g., *Dukhin and Shilov*, 2002]:

$$Du \equiv \frac{\sigma'_S}{\sigma_f}. \quad (11)$$

The approach of *Revil et al.* [1998] is slightly more complicated than equations (9) and (10) above because it accounts

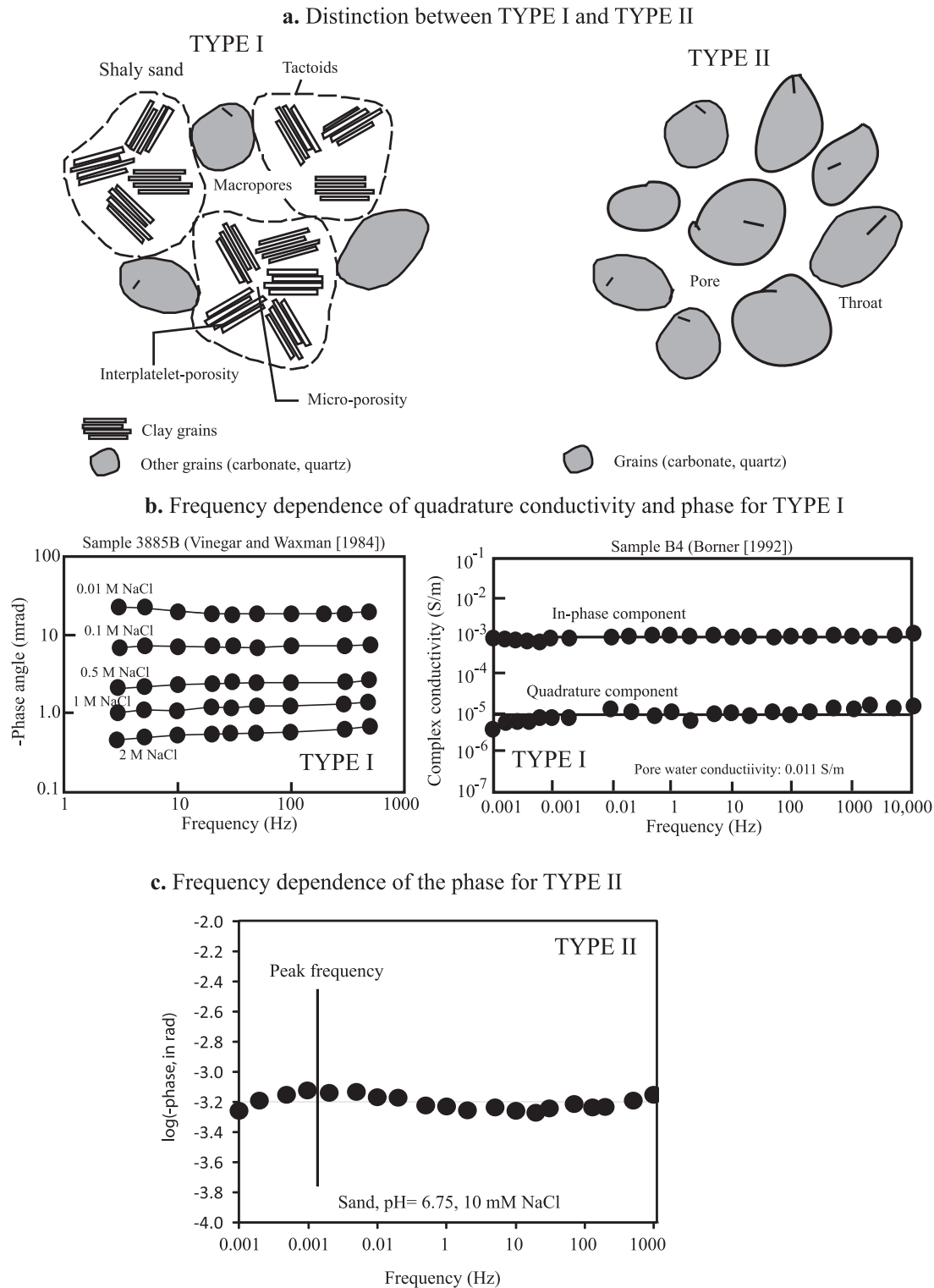


Figure 2. Shaly sand. (a) Microstructure with the distinction between media characterized by a broad range of polarization length scales like clayey sands (type I) and clean sand (type II) for which the pore volume is characterized by a succession of pores and throats. (b) Example of phase spectra for a sample of the database of *Vinegar and Waxman* [1984] (left) and the in-phase and quadrature conductivity spectra for a sample of the database of *Börner* [1992] (right). In both cases, the weak dependence of the phase or quadrature conductivity is the result of the broad distribution of polarization length scales. (c) The phase for type II materials (clean sand) shows a peak frequency.

for the different tortuosities felt by the anions and the cations during their electromigration in the pore space and along the mineral surfaces. However the difference between the *Revil et al.* [1998] model and equations (9) and (10) is rather small and will be ignored below.

[17] At low and high salinities, the asymptotic behaviors of equations (9) and (10) are

$$\sigma' = \sigma'_S, \text{ Du} \gg 1, \quad (12)$$

$$\sigma' = \frac{1}{F} [\sigma_f + m(F-1)\sigma'_S], \text{ Du} \ll 1, \quad (13)$$

respectively. Equation (13) is obtained from equation (7) by using binomial expansion [see *de Lima and Sharma*, 1992] and if equation (10) is used instead of equation (7), a similar expression is found with a value of cementation exponent $m = 2$ in the surface conductivity term [see *Niwas et al.*, 2007].

[18] I need now to express the surface conductivity term in equations (12) and (13) as a function of the mobility and density of the counterions. The counterions are defined as being the ions of opposite sign to the sign of the charge of the mineral surface. They are populating the electrical double layer to shield the charge of the clay particles. The counterions are located both in the Stern layer and in the diffuse layer (Figure 1). Triple-layer models (TLMs) have been developed to describe the charging properties of clays. The TLM of *Leroy and Revil* [2009] consider that the Stern layer is essentially associated with the edge of the clay particles. I will follow this idea below. I use f ($0 \leq f \leq 1$) to denote the relative fraction of counterions in the Stern layer ($f = 1$ if all the counterions are located in the Stern layer and $f = 0$ if all the counterions are located in the diffuse layer). This parameter (also called the partition coefficient hereinafter) will be of crucial importance to model the complex conductivity of shaly sands. The surface conductivity can be related to the excess charge density Q_V . This excess charge density represents the excess of electrical charges per unit pore volume of the porous material (in C m^{-3}). The relationship between the in-phase component of the surface conductivity and the excess charge density is [Kan and Sen, 1987; *Revil et al.*, 1998; *Brovelli and Cassiani*, 2011]

$$\sigma'_S = \frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \beta_S Q_V, \quad (14)$$

where β_S represents an effective (or apparent) surface mobility for the counterions of the electrical double layer. As noted by *Vinegar and Waxman* [1984], the excess of charge per unit pore volume Q_V can be determined (1) intrusively through cation exchange capacity (CEC) measurements obtained by potentiometric titration methods [e.g., *Meier and Kahr*, 1999] or (2) nonintrusively through membrane potential measurements [Thomas, 1976]. The relationship between the excess of charge per unit pore volume and the CEC is given by

$$Q_V = \rho_g \left(\frac{1-\phi}{\phi} \right) \text{CEC}, \quad (15)$$

where ρ_g is the grain density (in kg m^{-3}) and the CEC denotes the cation exchange capacity (in meq g^{-1} or in C kg^{-1} in the international system of units) of the clay minerals (an arithmetic average can be used for a mixture of clay minerals or mixed layer clays [see *Meier and Kahr*, 1999, and references therein]). The CEC represents the amount of cations exchange sites on the mineral surface per unit mass of the minerals. This parameters is of crucial importance in many properties of porous media like defining sorption isotherms for clay-rich materials [Woodruff and *Revil*, 2011].

[19] Equation (14) does not account for the different contributions from the Stern and diffuse layers. To account for these contributions, the surface conductivity needs to be written explicitly as a function of the mobility of the counterions in the Stern and diffuse layers as well as the relative fractions f and $(1-f)$. This yields

$$\sigma'_S = \frac{2}{3} \left(\frac{\phi}{1-\phi} \right) (\beta_S^{(+)} f Q_V + \beta_{(+)} (1-f) Q_V), \quad (16)$$

where $\beta_S^{(+)}$ denotes the mobility of the counterions in the Stern layer and $\beta_{(+)}$ represents the mobility of the cation in the pore water. I will consider below that the pore water solution is an electrolyte of NaCl. Comparing equations (14) and (16), the effective (or apparent) surface mobility β_S is related to the mobility of the cations in the Stern and diffuse layers by

$$\beta_S = \beta_S^{(+)} f + \beta_{(+)} (1-f). \quad (17)$$

Revil et al. [1998] found that for NaCl solutions (the counterions are Na^+ for pH values in the range 5 to 7) $\beta_S = \beta_S^{(+)} / 10$. As the partition coefficient is $f \approx 0.9$ (as discussed in section 2.3 in more detail [see also *Leroy and Revil*, 2009; *Revil and Florsch*, 2010]), this means that

$$\beta_S^{(+)} \ll \beta_{(+)} \Rightarrow \begin{cases} \beta_S \approx \beta_{(+)} (1-f) \\ \sigma'_S \approx \frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \beta_S Q_V \end{cases} \quad (18)$$

Therefore the mobility of the counterions in the Stern layer is expected to be much smaller than the mobility of the counterions in the diffuse layer. I investigate below how much smaller. This idea is consistent with the high density of counterions in the Stern layer of clay minerals. Indeed, this high density should lead to a smaller mobility of the counterions by comparison with its value in the bulk pore water. I will show in sections 2.2 and 2.3 that independent observations confirm this point of view and can be, at the same time, reconciled with previous modeling efforts.

[20] From equations (13) and (16), at high salinities, the in-phase component of the surface conductivity is written as

$$\sigma' = \frac{1}{F} [\sigma_f + A(\phi, m) \beta_S Q_V], \text{ Du} \ll 1, \quad (19)$$

$$A(\phi, m) = m(F-1) \frac{2}{3} \left(\frac{\phi}{1-\phi} \right), \quad (20)$$

where m is the same cementation exponent arising in Archie's law. Equation (19) is consistent with the high-salinity behavior described by *Sen and Goode* [1992]. Note at high porosities and for spherical grains ($m = 1.5$), the asymptotic behavior of equation (20) yields $A(\phi, m) \rightarrow m$ providing a rational for the *Waxman and Smits* [1968] equation, an approximation valid only at high salinities and high porosities.

[21] In section 3 I will use mainly the databases of *Vinegar and Waxman* [1984] and *Börner* [1992] to test my model. As discussed below, these databases are showing that the quadrature conductivity is independent of the frequency for a broad ranges of frequencies. As the conductivity data of *Vinegar and Waxman* [1984] are reported as a function of the salinity and temperature, I need to convert these data into pore water conductivities. For a NaCl solution, the conductivity of the pore water is given very accurately as a function of the salinity and temperature by the empirical formula obtained by *Sen and Goode* [1992]:

$$\sigma_f = (5.6 + 0.27T - 1.5 \cdot 10^{-4}T^2)C_f - \left(\frac{2.36 + 0.099T}{1.0 + 0.214C_f} \right) (C_f)^{\frac{3}{2}}, \quad (21)$$

where C_f is the salinity (expressed in Mol L⁻¹) and T the temperature (in °C).

2.2. Quadrature Conductivity

[22] As explained by *Vinegar and Waxman* [1984], the quadrature conductivity is controlled by two processes working in series, the weakest one dominates therefore the response (see Figure 2b). These two processes are namely the Stern layer polarization [see *Leroy et al.*, 2008] and the membrane polarization [Marshall and Madden, 1959; Titov et al., 2002]. The rational for these two processes working in series is explained by *Vinegar and Waxman* [1984] and will be investigated in more detail in a future publication.

[23] The quadrature conductivity is dominated by the Stern layer contribution, which is generally much smaller than the membrane polarization except at very high salinities for which the membrane polarization is very small and therefore may control the overall response of the material [Vinegar and Waxman, 1984]. The second point is that the quadrature conductivity is frequency dependent but this frequency dependence is weak because of the very broad distribution of polarization length scales (see discussion by *Vinegar and Waxman* [1984] and their Figures 4 and 5). This implies that I can neglect the frequency dependence of the quadrature conductivity as a first-order approximation. This is not the case for well-sorted sands for which a clear relaxation peak is usually observed in the phase lag and in the quadrature conductivity. A complete theory with membrane polarization will be kept for a future work and I will focus below on the Stern layer polarization model. This limits however the application of my model to clayey sands and sandstones.

[24] The Stern polarization model is based on the assumption that the Stern layer is discontinuous between the grains. This assumption is consistent with transport models for ions through clayey material [Revil et al., 2011, and

references therein] and with compaction models by pressure solution for sandstones [Revil, 2001].

[25] For silica sands, neglecting the frequency dependence of the quadrature conductivity, the quadrature conductivity is related to the quadrature component of the surface conductivity by [Revil and Florsch, 2010]

$$\sigma'' = -\frac{F-1}{F} \sigma_s'' \approx -\sigma_s''. \quad (22)$$

The last term of equation (22) is obtained with $(F-1)/F \approx 1$ because $F \gg 1$ (typically $F > 10$ for shaly and clayey sandstones; see Tables 1 and 2). This assumption is however not crucial to the model and we can carry along the term $(F-1)/F$ further into the modeling if needed. *Revil and Florsch* [2010] developed the following formula for the quadrature conductivity of a clean sand with a single grain diameter d_0 (again neglecting the frequency-dependent term):

$$\sigma'' = -\frac{F-1}{F} \left(\frac{4\Sigma_s}{d_0} \right) \approx -\left(\frac{4\Sigma_s}{d_0} \right), \quad (23)$$

where Σ_s is the specific surface conductivity associated with the Stern layer (in S). If the grains have different grain diameters, the particle size distribution (PSD) is described by a (normalized) probability distribution $f(D)$ (D is the grain diameter). Equation (23) can be generalized to [Revil and Florsch, 2010]

$$\sigma'' = -4E_h \Sigma_s, \quad (24)$$

$$E_h = \int_0^{+\infty} f(D) d \ln D. \quad (25)$$

Therefore, the mean grain size needs to be replaced by the expectation of the moment of order (-1) of the PSD. I know from the conductivity theories developed by *Revil et al.* [1998] and *Revil and Florsch* [2010] that the following equivalence applies for the surface (interfacial) electrical conductivity [e.g., *Schmutz et al.*, 2010]:

$$\frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \beta_{(+)}^s f Q_V \iff 4E_h \Sigma_s. \quad (26)$$

There is therefore a bridge between expressing the surface conductivity as a function of the specific surface conductivity Σ_s (and the moment of order (-1) of the PSD) and as a function of the excess charge density per unit pore volume Q_V (or alternatively the CEC) (see *Brovelli and Cassiani* [2011] on the same topic).

[26] From equations (24) and (26), the quadrature conductivity model of *Revil and Florsch* [2010] can be written as

$$\sigma'' \approx -\frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \beta_{(+)}^s f Q_V. \quad (27)$$

Like I have done above for the in-phase surface conductivity model with β_s , I introduce now an apparent

Table 1. Properties of the Sandstones From the *Vinegar and Waxman* [1984] Database

Sample	Mineralogy ^a	F^b	$\sigma'_s (\times 10^{-4} \text{ S m}^{-1})$	ϕ^d	$Q_V^e (\text{meq mL}^{-1})$	f_M^f	CEC ^g (C kg^{-1})	k^h (mD)
3477	K, C	15.1 ± 0.2	18.0 ± 2.0	0.201	0.026	0.92	237.74	833
3336A	K	23.1 ± 0.5	13.7 ± 2.2	0.205	0.042	0.96	393.65	71.7
3478	K, C	19.0 ± 0.6	12.6 ± 3.2	0.181	0.052	0.97	417.70	78.2
101	K, I	16.5 ± 0.3	26.2 ± 2.7	0.210	0.055	0.95	531.40	-
102	K, I	16.2 ± 0.4	7.40 ± 2.4	0.205	0.064	0.99	599.84	-
CZ10	C	18.0 ± 0.3	47.0 ± 3.0	0.233	0.070	0.93	772.91	116
3833A	C	13.9 ± 0.6	38.0 ± 8.0	0.224	0.110	0.96	1154.1	118
3126B	I, S	11.8 ± 0.3	64.0 ± 6.0	0.249	0.120	0.95	1446.1	468
3847A	I, C	48.4 ± 1.3	18.5 ± 1.9	0.126	0.144	0.97	754.56	0.66
3283A	K, C, I	19.5 ± 1.0	27.2 ± 7.0	0.186	0.150	0.98	1245.8	40
3885B	I, C	29.0 ± 1.5	32.0 ± 7.0	0.204	0.180	0.98	1676.7	4.95
3972E	S, M, I	21.9 ± 0.4	27.6 ± 2.4	0.187	0.185	0.98	1546.7	407
3258A	I	51.8 ± 3.0	62.0 ± 9.0	0.128	0.379	0.97	2022.1	0.61
3891A	C, I	42.1 ± 1.3	173 ± 9	0.192	0.385	0.94	3325.2	0.26
3308A	S, C	13.7 ± 0.7	320 ± 35	0.270	0.409	0.94	5498.4	5
3323F	S, C	15.1 ± 0.4	749 ± 48	0.288	0.690	0.92	10145	3.1
3324A	S, C	32.7 ± 1.1	670 ± 34	0.220	0.646	0.89	6622.6	1.48
3323E	S, C	15.8 ± 0.6	798 ± 62	0.281	0.690	0.91	9801.6	1.26
3324B	S, C	32.0 ± 1.0	650 ± 30	0.204	0.842	0.91	7843.3	1.14
3306F	S, C	50.0 ± 5.0	538 ± 95	0.171	0.950	0.92	7122.6	0.70

^aFrom *Vinegar and Waxman* [1984, Table 1]. K, kaolinite; C, chlorite; I, illite; S, smectite.^bFrom the fit of the in-phase conductivity data using the differential effective medium theory (DEM).^cFrom the fit of the in-phase conductivity data using the DEM.^dFrom *Vinegar and Waxman* [1984].^eFrom *Vinegar and Waxman* [1984].^fFrom the surface conductivity, porosity, and Q_V .^gCation exchange capacity. From Q_V and porosity assuming a mass density of 2650 kg m^{-3} .^hMeasured brine permeability from *Vinegar and Waxman* [1984].

mobility, λ (in $\text{m}^2 \text{ s}^{-1} \text{ V}^{-1}$, using the same notation as *Vinegar and Waxman* [1984]), for the surface quadrature conductivity as

$$\sigma'' \approx -\frac{2}{3} \left(\frac{\phi}{1-\phi} \right) \lambda Q_V \quad (28)$$

$$\lambda \equiv \beta_{(+)}^S f. \quad (29)$$

I need now to find an expression for the partition coefficient f , which is entering in both the surface conductivity and the

quadrature conductivity contributions. This is done in section 2.3.

2.3. A Model for the Partition Coefficient

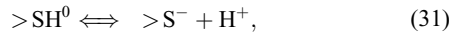
[27] As mentioned above, the last component of the model to investigate is the salinity dependence of the partition coefficient f . My goal is to develop a simplified, first-order solution of the TLM sorption problem of *Leroy and Revil* [2009]. I consider that the sorption of sodium in the Stern layer of kaolinite (illite and smectite are discussed below) and the dissociation of protons can be represented by the following reactions [e.g., *Leroy et al.*, 2007]

Table 2. Properties of the Sandstones From the *Börner* [1992] Database

Sample	Type ^a	F	$\sigma'_s (\times 10^{-4} \text{ S m}^{-1})$	ϕ	$S_{por} (\mu\text{m}^{-1})$	$S_{sp} (10^3 \text{ m}^2 \text{ kg}^{-1})$	CEC ^c (C kg^{-1})	k (mD)
E6	Saxony	21.7^d	$1.29\text{e-}2$	0.1980	31.1	2.96	1125	-
E7	Saxony	11.5^d	$1.70\text{e-}2$	0.2762	13.2	1.95	741	157
E10	Saxony	17.0	$1.85\text{e-}2$	0.2238	41.6	4.64	1763	5.4
E12	Saxony	11.0	$8.61\text{e-}3$	0.2989	5.5	0.748	284	1490
E14	Saxony	11.1	$9.58\text{e-}3$	0.2403	10.7	1.31	498	1990
E17	Saxony	25.8^d	$3.74\text{e-}3$	0.1806	1.44	0.122	46	680
B2	Bentheim	15.6	$1.57\text{e-}2$	0.1943	2.15	0.198	74	1880
B4	Bentheim	11.7^b	$7.63\text{e-}3$	0.2315	2.22	0.261	99	-
R1	Altmark	150^e	$3.96\text{e-}2$	0.0463	43.5	0.809	307	-
R3	Altmark	45.1	$1.07\text{e-}2$	0.0970	24.7	1.01	384	0.52
F1	Fontainebleau	12.6	$7.62\text{e-}3$	0.2143	0.146	0.0153	6	-
F3	Fontainebleau	118	$7.19\text{e-}4$	0.0876	0.126	0.0036	1.4	138

^aThe Saxony sandstone contains quartz, feldspar, and kaolinite. The Bentheim sandstone contains silica (95%) and some kaolinite and dickite. The Altmark sandstone contains silica with a coating of smectite and illite and kaolinite and illite as cements. The Fontainebleau sandstone is nearly pure silica (99.8%).

^bUsing the porosity and $m = 1.68$ (from sample B2).^cUsing $\text{CEC} = Q_S S_{sp}$, with $Q_S = 0.32 \text{ C m}^{-2}$.^dUsing the porosity and $m = 1.90$ (from the other Saxony samples).^eUsing the porosity and $m = 1.63$ (from sample R3).



where $>S$ refers the surface sites attached to the crystalline framework, H^0 are protons (which are assumed to be immobile) while weakly sorbed Na^+ are considered to be mobile in the Stern layer, eventually with a very small mobility. For illite and smectite, only equation (30) should be considered. I will consider the protons as immobile for clays but they seems to be mobile for silica as shown recently by *Skold and Revil* [2011]. Another component to consider is the charge associated with isomorphous substitutions in the crystalline framework. This involves mainly the substitution of Al^{3+} for Si^{4+} in the tetrahedral sheet of the clay particles and Mg^{2+} or Fe^{2+} for Al^{3+} in the octahedral sheet of the clay particles. These substitutions are important for illite and smectite but not for kaolinite. In agreement with the work of *Leroy and Revil* [2009], I will consider that the sites associated with the isomorphous substitutions (essentially the basal planes) are not associated with the formation of a Stern layer.

[28] To find a simplified sorption isotherm, I use the following equilibrium (constitutive) relationships corresponding to equations (30) and (31):

$$K_{Na} = \frac{\Gamma_{SNa}^0 [H^+]^0}{\Gamma_{SH}^0 [Na^+]^0}, \quad (32)$$

$$K_H = \frac{\Gamma_{S^-}^0 [H^+]^0}{\Gamma_{SH}^0}, \quad (33)$$

and a conservation equation for the surface species,

$$\Gamma_S^0 = \Gamma_{SNa}^0 + \Gamma_{SH}^0 + \Gamma_{S^-}^0 + \Gamma_X^0, \quad (34)$$

where Γ_S^0 denotes the (known) total surface site density (including the charge associated with isomorphous substitutions in the crystalline framework), Γ_{SNa}^0 , $\Gamma_{S^-}^0$, and Γ_{SH}^0 represent the surface charge density of the sites $>S^-Na^+$, $>S^-$, and $>SH^0$, respectively, and Γ_X^0 represents the number of equivalent sites corresponding to isomorphous substitutions (all expressed in sites m^{-2}). The value of the equilibrium constant is approximately $K_{Na} = 7.7 \times 10^{-2}$ [*Leroy et al.*, 2007], and the value of the value of K_H is between 10^{-10} for aluminol surface sites and 8×10^{-6} for silanol sites [*Leroy and Revil*, 2009]. The continuity equation (34) implies that the sites corresponding to the isomorphous substitutions are not attached with sorption in a Stern layer because they do not correspond to true surface sites. I simplify the notations using $pH = -\log_{10} [H^+]$ and $[Na^+] = C_f$, the salinity. The resolution of equations (32) to (34) is straightforward and yields the following sorption isotherm:

$$\Gamma_{SNa}^0 = (\Gamma_S^0 - \Gamma_X^0) \frac{K_{Na} C_f}{10^{-pH} + K_{Na} C_f + K_H}. \quad (35)$$

The partition coefficient is defined by

$$f \equiv \frac{\Gamma_{SNa}^0}{\Gamma_{S^-}^0 + \Gamma_X^0 + \Gamma_{SNa}^0}. \quad (36)$$

This means that the partition coefficient is defined as the surface concentration of counterions in the Stern layer divided by the total surface concentration of counterions in both the diffuse and the Stern layer (the negative charges $\Gamma_{S^-}^0 + \Gamma_X^0$ are compensated in the diffuse layer). This yields

$$f = f_M \left(\frac{K_{Na} C_f}{10^{-pH} (1 - f_M) + K_{Na} C_f + K_H} \right), \quad (37)$$

where the maximum value of the partition coefficient f_M (reached at high salinity) is defined from equations (35) and (36) as

$$f_M = 1 - \frac{\Gamma_X^0}{\Gamma_S^0}. \quad (38)$$

For illite and smectite, the activity of the silanol and aluminol surface groups can be neglected and therefore K_H can be neglected in equation (37). The maximum partition coefficients are shown in Figure 3 from the work of *Leroy and Revil* [2009]: $f_M(\text{kaolinite}) = 0.98$ (and therefore $\Gamma_X^0 \approx 0.02$ from equation (38), a very small value as expected from crystallographic considerations), $f_M(\text{illite}) = 0.90$ (and therefore $\Gamma_X^0 = 0.1 \Gamma_S^0$), and $f_M(\text{smectite}) = 0.85$ (and therefore $\Gamma_X^0 = 0.15 \Gamma_S^0$, which is consistent with the values reported by *Tertre et al.* [2011]).

[29] From equation (29) and equations (37) and (38), the apparent mobility entering into the expression of the quadrature conductivity can be expressed as follows:

$$\lambda = \lambda_M \left(\frac{K_{Na} C_f}{10^{-pH} (1 - f_M) + K_{Na} C_f + K_H} \right), \quad (39)$$

$$\lambda_M = \beta_{(+)}^S f_M, \quad (40)$$

where λ_M denotes the maximum value of the apparent mobility reached at high salinity. Again for smectite and illite, K_H can be neglected in equation (39). Equation (39) provides an explicit dependence of the apparent mobility of the counterions with the salinity. At high salinity, the model predicts that the apparent mobility λ is independent of the salinity (except if I account for membrane polarization effects) and at low salinity, the apparent mobility increases with the salinity. Therefore the Stern polarization model implies that there is an increase of the polarization with the salinity because the density of counterions in the Stern layer increases with the salinity. I will show in section 3 that this prediction is consistent with the available experimental data.

3. Validation of the Model

[30] I test ϕ below several predictions of the model with respect to the dependence of the quadrature conductivity with the variables entering into the model. These variables

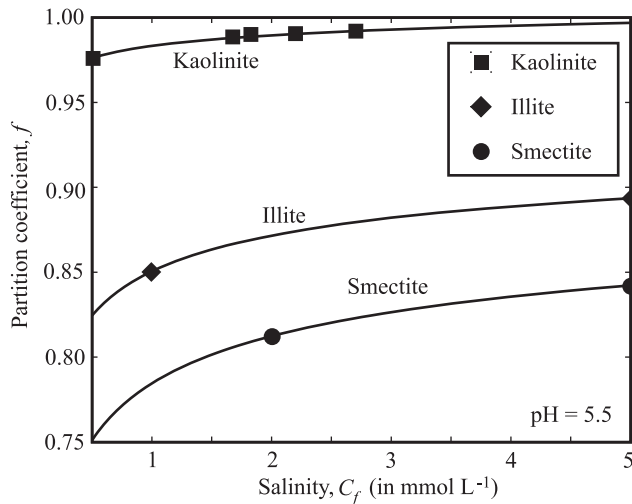


Figure 3. Partition coefficients for the three main types of clay minerals (chlorite has properties very similar to kaolinite). The lines correspond to the prediction of the triple-layer model (TLM) calculations [from Leroy and Revil, 2009] and the symbols correspond to estimates using the spectral induced polarization model of Leroy and Revil [2009] and complex conductivity measurements.

include textural variables (porosity, surface area), electrochemical variables (CEC or charge per unit surface area), and environmental variables (salinity and temperature). I use essentially two databases to test the model: one from Vinegar and Waxman [1984] (measured at 30 Hz and 25°C) and the second from Börner [1992] (measured at 1 Hz and room temperature) because of the availability of all the data. These databases corresponds to clayey sands and sandstones.

3.1. Quadrature Conductivity Versus Excess Charge Density

[31] The salinity dependence of the in-phase conductivity is analyzed in Figure 4. As all the data are above the iso-conductivity point (corresponding to $Du = 1$), I first use equation (10) and I minimize the data misfit (RMS) error between the measured (in-phase) conductivity and the in-phase conductivity predicted of the model. This first step yields the value of the formation factor F and the value of the in-phase surface conductivity σ'_s reported in Figure 4a for sample 3847A from Vinegar and Waxman [1984] (see Table 1) and in Figure 4b for sample B2 from Börner [1992] (see Table 2).

[32] The second step of the analysis is to compute the partition coefficient f_M using equation (18) and the value of the in-phase surface conductivity. For sample 3847A, using the value of the connected porosity ($\phi = 0.126$) and the value of the excess of charge per unit pore volume ($Q_V = 0.144 \text{ meq mL}^{-1} = 13.9 \times 10^6 \text{ C m}^{-3}$), and using $\beta_{(+)}(\text{Na}^+, 25^\circ\text{C}) = 5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ and the value of the in-phase surface conductivity from the first step, I obtain the following value for the partition coefficient $f_M = 0.97$ using the value of the surface conductivity. For sample B2, the same analysis yields $f_M = 0.986$.

[33] The third step is to predict the value of the quadrature conductivity, neglecting at first its salinity dependence, which will be investigated only later on in this paper (see Figure 4b). So I use equations (28) and (29) with $f = f_M$. For sample 3847A, using $f_M = 0.97$ and the value of the mobility of the counterions in the Stern layer $\beta_{(+)}^S(\text{Na}^+, 25^\circ\text{C}) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, I obtain $\sigma'' = 1.94 \times 10^{-4} \text{ S m}^{-1}$, which is in good agreement with the measured values (see Figure 4a). For sample B2, using $f_M = 0.986$, $\beta_{(+)}^S(\text{Na}^+, 25^\circ\text{C}) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ at 25°C , I obtain $\sigma'' = 1.93 \times 10^{-5} \text{ S m}^{-1}$, also in good agreement with the measured values. The salinity dependence of the quadrature conductivity will be investigated in section 3.8.

[34] I use equation (10) to fit all the in-phase conductivity data reported by Vinegar and Waxman [1984] in order to determine the value of the (intrinsic) formation factor F and the value of the mean surface conductivity σ_s for each sample. These values are reported in Table 1 for all the samples of this database. Figure 5 shows the formation factors determined from the DEM model versus the formation factors given by Vinegar and Waxman [1984] for their database using the Waxman and Smits [1968] (WS) linear model. Both estimates are in fairly good agreement with each other, the formation factor estimated from the DEM being slightly smaller than the estimates provided by the WS model.

[35] The surface conductivities reported in Table 1 are shown as a function of the excess of charge density Q_V in Figure 6. According to equations (16) and (18), the slope of the trend is given by $s = (2/3)\beta_{(+)}(1 - f) = (2.8 \pm 0.2) \times 10^{-9} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. Using the mobility of sodium in the pore water, this yields $f_M = 0.92$ as an average value for the partition coefficient for the complete database. Using a broader database, Revil *et al.* [1998] obtained $s = (5.1 \pm 0.2) \times 10^{-9} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. This yields $f_M = 0.85$ as an average value for the partition coefficient. Usually these estimates are biased by the samples with the larger CEC (usually with smectite) and they are therefore more representative of smectite than of kaolinite. These estimates are therefore consistent with those reported in Figure 3 for the partition coefficient.

[36] I can also compute the partition coefficient f_M for each sample using

$$f_M = 1 - \frac{3}{2} \left(\frac{1 - \phi}{\phi} \right) \left(\frac{\sigma'_s}{\beta_{(+)} Q_V} \right), \quad (41)$$

where σ'_s denotes the surface conductivity determined from the in-phase conductivity data and ϕ and Q_V are independently measured in the database of Vinegar and Waxman [1984]. The computed partition coefficient are reported as a function of the CEC in Figure 7. I observe that small values of the CEC are associated with large values of $f_M > 0.95$, typical of kaolinite) while low values of f_M (in the range 0.87–0.95) are associated with high values of the CEC, typical of illite and especially smectite. This is again in agreement with the trends shown in Figure 3. These data can be fitted by $f_M = f_M(K) - 9.0 \times 10^{-6} \text{ CEC}$ where the CEC is expressed in C kg^{-1} and $f_M(K) = 0.98$ represents the value of the partition coefficient for kaolinite.

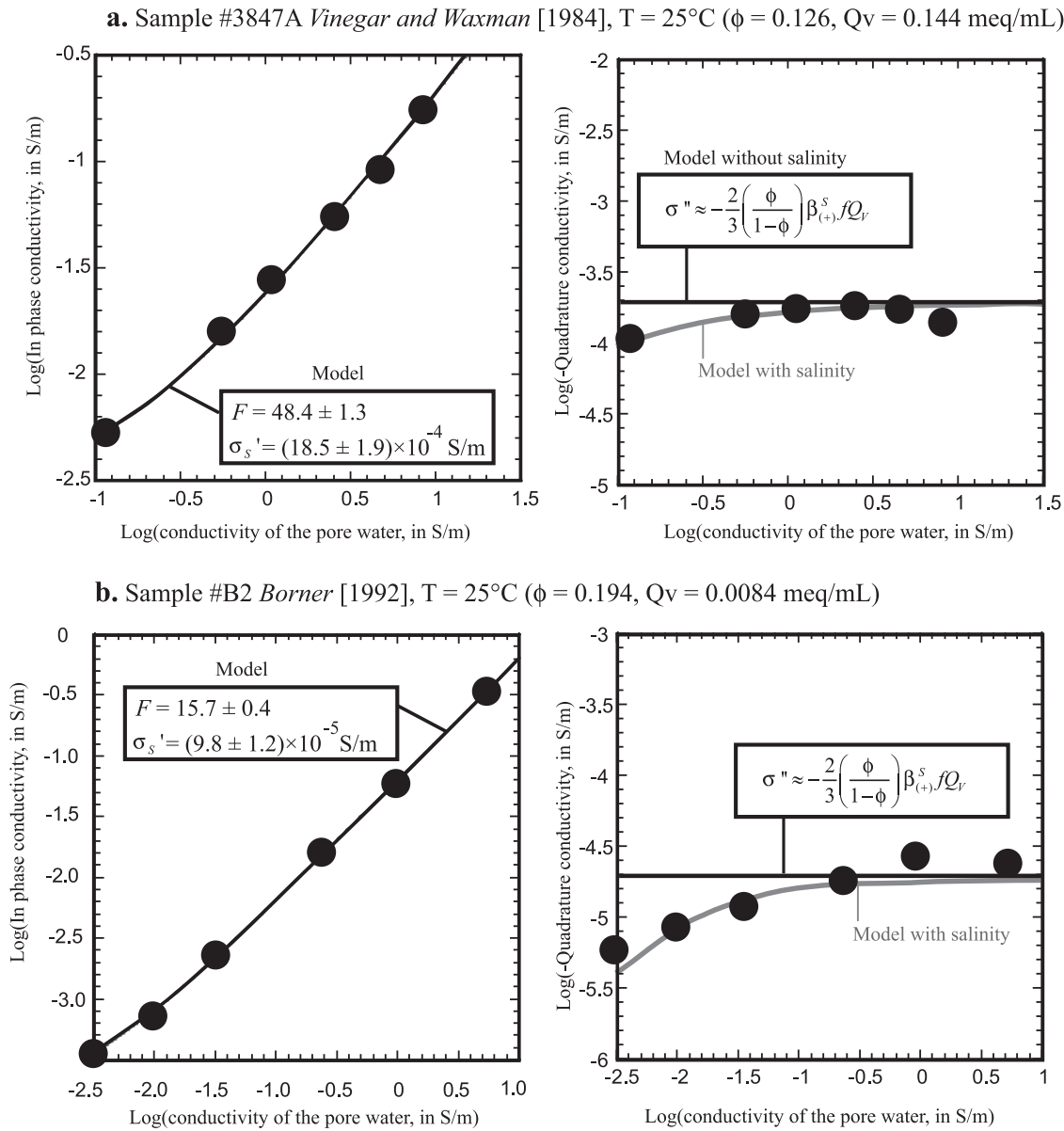


Figure 4. (a) Fit of the experimental data for a sample from the *Vinegar and Waxman* [1984] database. Note that the model captures well the nonlinear curvature of the in-phase conductivity data at low salinities. Note that all the data are above the isoconductivity point for which the conductivity of the sample is equal to the conductivity of the pore water and $Du = 1$. In the right plot, the information gathered by fitting the in-phase conductivity data are used to predict the quadrature conductivity (at 30 Hz, 25°) without using, at this stage, the salinity dependence of the partition coefficient. There is no flush factor in the model. The errors bars are smaller than the size of the symbols. (b) The same for one core sample from the database of *Börner* [1992].

[37] In Figure 8, I use these values of the partition coefficient (reported in Table 1) to analyze the relationship between the surface conductivity and the quadrature conductivity. Such an analysis is important to see if I can predict the surface conductivity from the quadrature conductivity. When the surface conductivity is normalized with respect to the value of the partition coefficient, I find a high correlation between the quadrature conductivity and this normalized surface conductivity (Figure 8). Note that a good correlation is also obtained by taking an average

value of the partition coefficient (not shown here). That said, if one wants to apply this relationship to field data, a rough idea of the partition coefficient is required in the form $f_M/(1 - f_M)$ and the relationship determined above, $f_M = f_M(K) - 9.0 \times 10^{-6}$ CEC, can be used for that purpose.

[38] In Figure 9a, I plot the normalized quadrature conductivities $[(1 - \phi)/\phi]\sigma''$ as a function of the excess charge density Q_v at four distinct salinities (I did not choose the very high salinity 2 Mol L^{-1} because I suspect that

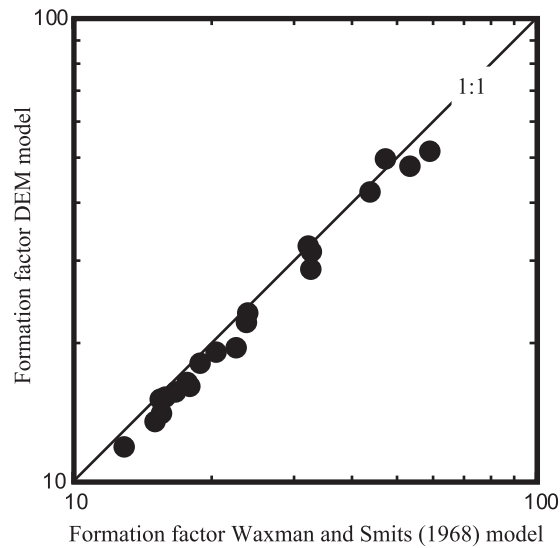


Figure 5. Comparison between the formation factor obtained from the fit of the data of *Vinegar and Waxman* [1984] using the differential effective medium (DEM) model and using the *Waxman and Smits* [1968] (WS) model. Note that the formation factors obtained using the DEM model are usually slightly smaller than those obtained with the WS model. The uncertainties are inside the size of the symbols.

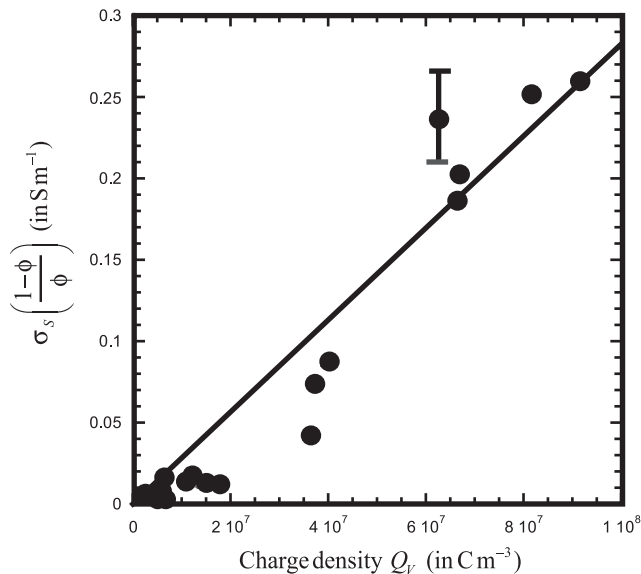


Figure 6. Comparison between the normalized (in-phase) surface conductivity (in S m^{-1}) and the excess charge density Q_v . The slope of the linear trend yields $s = (2/3)\beta_{(+)}(1 - f) = (2.82 \pm 0.15) \times 10^{-9} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$. The uncertainty given on one data point is representative of the data set and is determined from the uncertainty obtained from the RMS fit of the conductivity data using the DEM model. The scatter is coming from the mineralogical composition, which influences the values of the partition coefficient f . The error bar provides an idea of the order of magnitude of the uncertainties in the surface conductivity.

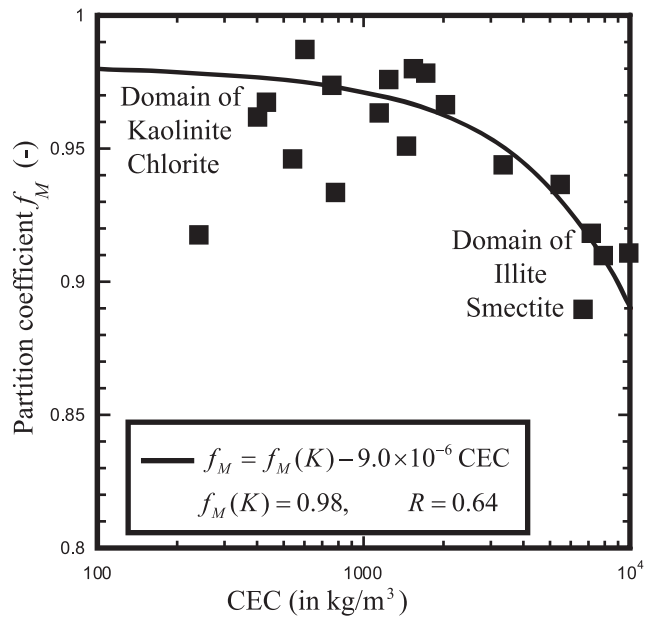


Figure 7. Value of the partition coefficient determined for each sample (from the surface conductivity) versus the cation exchange capacity (CEC) of the sample for the database of *Vinegar and Waxman* [1984]. At low CEC values, the partition coefficient is representative of kaolinite, while at high CEC the value of the partition coefficient is representative of illite and smectite. K stands for kaolinite. The main clay minerals of each sample are reported in Table 1.

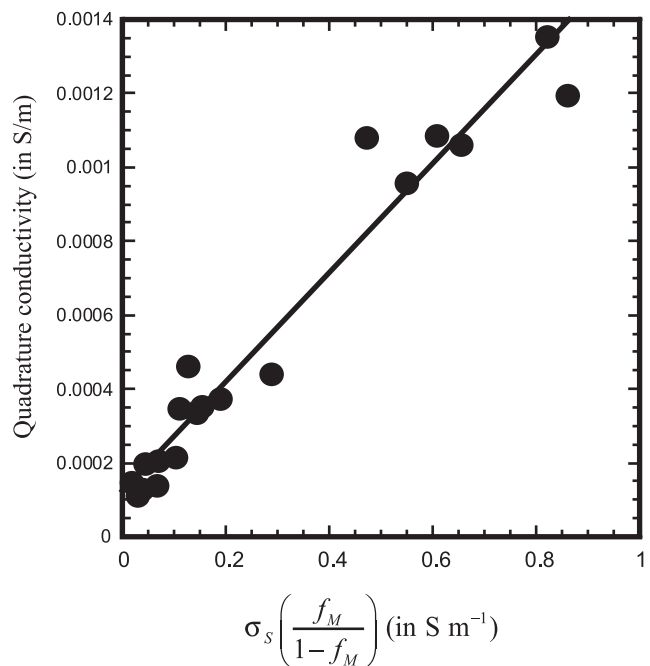


Figure 8. Quadrature conductivity versus scaled surface conductivity using the values of the partition coefficient reported in Table 1 for each sample. The slope of the linear trend ($R = 0.974$) is given by $s = \beta_{(+)}^S / \beta_{(+)} = 0.00148$ (dimensionless). This yields a counterion mobility in the Stern layer of $\beta_{(+)}^S = 0.8 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ at 25°C .

membrane polarization starts to be nonnegligible at this salinity). I see that the normalized quadrature conductivities and Q_V are highly correlated as predicted by the theory (see equation (27)) with a slope that is salinity dependent. At each salinity, I compute, for each sample, the apparent mobility λ according to

$$\lambda = -\frac{3}{2} \left(\frac{1-\phi}{\phi} \right) \frac{\sigma''}{Q_V}, \quad (42)$$

(from equation (28)). Then I compute the mean of λ for all the data, at each salinity, and I report these mean values of λ as a function of the salinity. The result is shown in Figure 9b. The result is fitted with equation (39). This fit provides the value of $\lambda_M = (1.32 \pm 0.03) \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ and the value of the sorption coefficient $K_{\text{Na}} = (3.0 \pm 0.4) \times 10^{-4}$ ($\log_{10} K_{\text{Na}} = -3.5$). The value of the sorption coefficient is lower than the value proposed by Leroy *et al.* [2007] for smectite ($K_{\text{Na}} = 7.7 \times 10^{-2}$). This difference could be due to the mineralogical composition of the samples investigated by Vinegar and Waxman [1984], as shown in Table 1. The value of λ_M can be used to infer the value of the intrinsic mobility of the counterions in the Stern layer. Taking $f_M = 0.90$ yields $\beta_{(+)}^S (\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, so a counterion

mobility approximately 350 times smaller than in the bulk pore water for Na^+ .

3.2. Influence of the CEC

[39] Combining the expressions of the quadrature conductivity, equation (27), and the relationship between the excess charge density and the CEC, equation (15), I obtain a direct relationship between the quadrature conductivity and the CEC:

$$\sigma'' \approx -\frac{2}{3} \beta_{(+)}^S f \rho_g \text{CEC}. \quad (43)$$

In other words, the quadrature conductivity is a direct measure of the cation exchange capacity of the porous material, which has huge implications in agriculture as discussed in section 4. Taking $\beta_{(+)}^S (\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $f = 0.9$, and $\rho_g = 2650 \text{ kg m}^{-3}$, I obtain $\sigma'' \approx -a \text{CEC}$ with $a \approx (2/3) \beta_{(+)}^S f \rho_g = 2.4 \times 10^{-7} \text{ S kg C}^{-1} \text{ m}^{-1}$. As shown in Figure 10a, the Stern layer polarization model is therefore able to explain the dependence of the quadrature conductivity with the cation exchange capacity of the material. In Figure 10b, I scale the values of the quadrature conductivity with the value of the partition coefficient f determined from

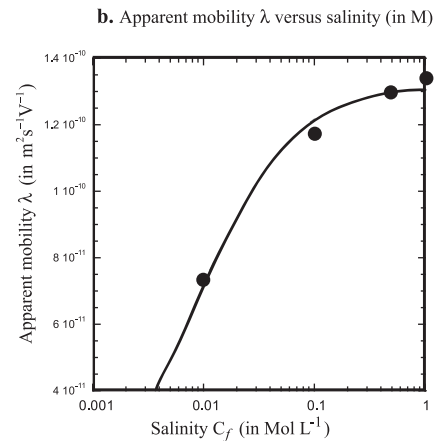
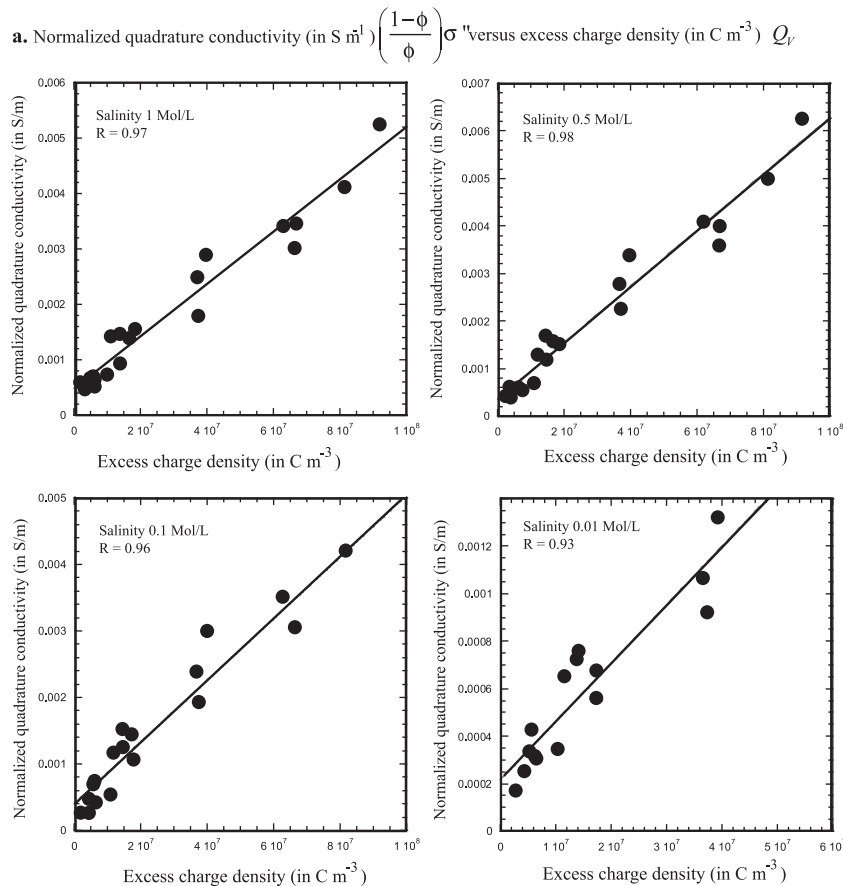


Figure 9. Quadrature conductivity versus the excess charge density and implication for the apparent mobility in the Stern layer. (a) Normalized quadrature conductivity versus excess charge density at four salinities. Note the excellent correlation for the linear trends. (b) Mean value of the apparent mobility obtained at each salinity versus the salinity of the pore water NaCl solutions. These data are fitted with the sorption isotherm described in section 2.3.

Quadrature conductivity versus CEC

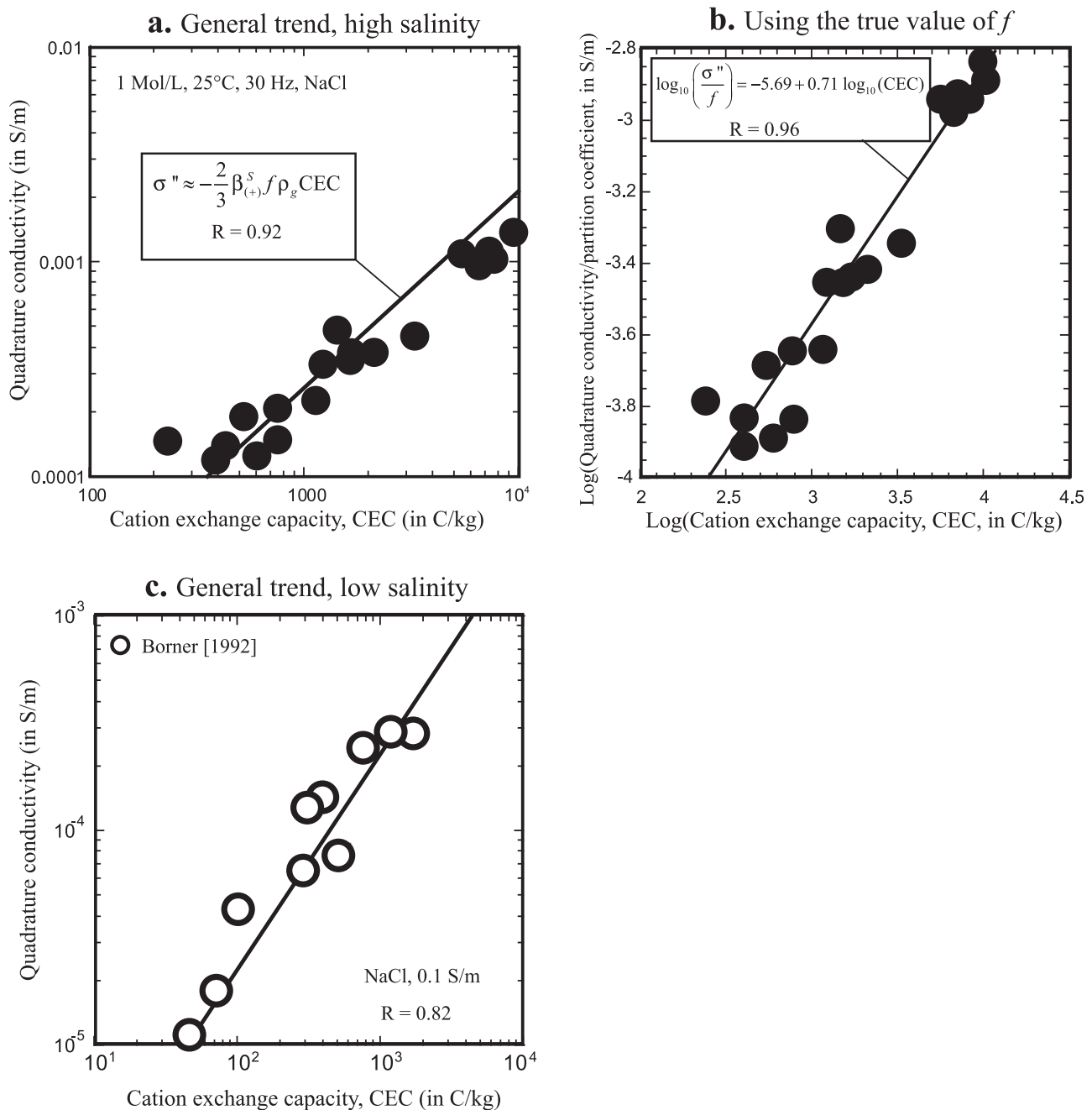


Figure 10. Relationship between the quadrature conductivity and the CEC (expressed in C kg^{-1}). (a and b) Data from the *Vinegar and Waxman* [1984] database. The model developed in the main text predicts a linear dependence between the quadrature conductivity and the CEC (lines), in agreement with the data. (c) Data from *Börner* [1992]. The line corresponds to the predicted response. The CEC is computed from the measured specific surface areas using $\text{CEC} = Q_s S_{sp}$, with the mean charge density for clay minerals Q_s being 2 elementary charges per nm^2 (0.32 C m^{-2}); see section 3.3.

the analysis of the in-phase conductivity. This improves the fit of the relationship between the scaled quadrature conductivity with the CEC. In Figure 10c, I used the database of *Börner* [1992] to test further the relationship between the quadrature conductivity and the CEC. This database does not contain CEC measurements, but the specific surface areas of

the core samples were measured. I use the relationship between the surface area and the CEC discussed below in section 3.3, equation (44), to convert the specific surface areas into CEC (see Table 2). Figure 10c is showing a strong correlation between the quadrature conductivity and the CEC. The slope of the trend is $(2.1 \pm 0.2) \times 10^{-7} \text{ S kg m}^{-1} \text{ C}^{-1}$, very

close to the slope given above ($2.4 \times 10^{-7} \text{ S kg C}^{-1} \text{ m}^{-1}$). Therefore, the model developed in this paper can clearly explain two published databases. The proportionality of the quadrature conductivity with the CEC is also supported by the data of *Garrouch and Sharma* [1994] (see their low-frequency dielectric dependence with the CEC, their Figures 3 and 4, for which one decade in the values of the CEC corresponds to one decade in the values of the apparent dielectric constant).

3.3. Influence of the Surface Area

[40] The charge density per unit surface area (in C m^{-2}) is defined as the ratio of the CEC by the specific surface area of the material S_{sp} (in $\text{m}^2 \text{ kg}^{-1}$)

$$Q_s = \frac{\text{CEC}}{S_{sp}}. \quad (44)$$

From Figure 11, the mean charge density for clay minerals is 2 elementary charges per nm^2 (0.32 C m^{-2} [see *Woodruff and Revil*, 2011]). Therefore the quadrature conductivity is also a measure of the specific surface area (in kg m^{-2}) of the material:

$$\sigma'' \approx -\frac{2}{3} \beta_{(+)}^s f \rho_g Q_s S_{sp}. \quad (45)$$

Taking $\beta_{(+)}^s (\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ at 25°C , $f = 0.90$, $Q_s = 0.32 \text{ C m}^{-2}$, $\rho_g = 2650 \text{ kg m}^{-3}$, we can write

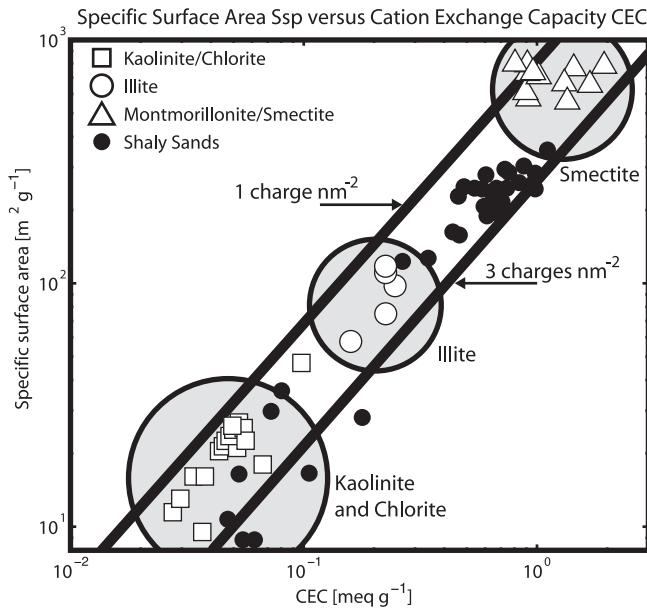


Figure 11. Specific surface area of clay minerals (in $\text{m}^2 \text{ g}^{-1}$) as a function of the CEC (in meq g^{-1} , where $1 \text{ meq g}^{-1} = 96,320 \text{ C kg}^{-1}$ in SI units) for various clay minerals. The ratio between the CEC and the specific surface area gives the equivalent total surface charge density of the mineral surface. Generalized regions corresponding to kaolinite, illite, and smectite are represented by shaded circles. Figure adapted from *Revil and Leroy* [2004]. The two lines correspond to 1–3 elementary charges per nm^2 . The average charge density is therefore 2 ± 1 elementary charges per nm^2 .

$\sigma'' \approx -a S_{sp}$ with $a = 7.6 \times 10^{-8} \text{ S kg m}^{-3}$. In Figure 12, I show the correlation between the quadrature conductivity and the specific surface area for the sandstones core samples investigated by *Börner* [1992]. There is a high correlation between the two parameters while the porosity of the core samples is inside a broad range (see Table 2). I find $\sigma'' \approx -a S_{sp}$ with $a = (8.0 \pm 0.9) \times 10^{-8} \text{ S kg m}^{-3}$. In other words, there is again a good agreement between the theory and the experimental data using this specific distinct data set.

[41] *Weller et al.* [2010] observed a linear relationship between the quadrature conductivity and the specific surface per unit pore volume at 1 Hz for 114 samples of sandstones and mix of sands and clays. The specific surface per unit pore volume can be related to the charge density Q_V and the surface charge density Q_s by

$$S_{por} = \frac{S}{V_p} = \frac{Q_V}{Q_s} \quad (46)$$

$$S_{por} = \rho_g \left(\frac{1 - \phi}{\phi} \right) S_{sp}. \quad (47)$$

Therefore the quadrature conductivity is given by

$$\sigma'' \approx -\frac{2}{3} \left(\frac{\phi}{1 - \phi} \right) \beta_{(+)}^s f Q_s S_{por}. \quad (48)$$

Clearly this relationship depends also on the porosity and the partition coefficient f , which explains the scatter in the data reported by *Weller et al.* [2010, Figure 2]. Taking average values of $\beta_{(+)}^s (\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $\phi = 0.30$, $Q_s = 0.32 \text{ C m}^{-2}$, and $f = 0.90$, I obtain $\sigma'' \approx -c_p S_{por}$ with $c_p = 12.3 \times 10^{-12} \text{ S}$ with S_{por} expressed in m^{-1} and σ'' expressed in S m^{-1} . *Weller et al.* [2010, Table 1 and Figure 2] obtained $c_p = 9.9 \times 10^{-12} \text{ S}$. To my knowledge, this is the first time that a model can explain the dependence of the quadrature conductivity of shaly sands with the specific surface per unit pore volume. That said, the relationship between the quadrature conductivity and the specific surface area should have less scatter because it does not depend on the porosity (compare equations (45) and (48)). Indeed, from equation (47), it is clear that S_{por} play a role similar to Q_V in equation (15) while S_{sp} plays a role similar to the CEC in equation (15). Figure 12, using again the data set of *Börner* [1992] comprising core samples with a broad porosity range (see Table 2), shows that there is a higher correlation between σ'' and S_{sp} than between σ'' and S_{por} . A similar observation was reported by *Slater and Glaser* [2003].

3.4. Critical Phase Angle at Very Low Salinities

[42] The model of *Vinegar and Waxman* [1984] predicts the existence of a limiting value of the phase lag at very low salinities. According to my model, at very low salinities ($Du \gg 1$), the in-phase conductivity is given by $\sigma' \approx \sigma'_s$ and therefore the phase lag reaches indeed a critical value given by

$$\lim_{Du \gg 1} \varphi = -\frac{\beta_{(+)}^s f}{\beta_{(+)}(1 - f)}. \quad (49)$$

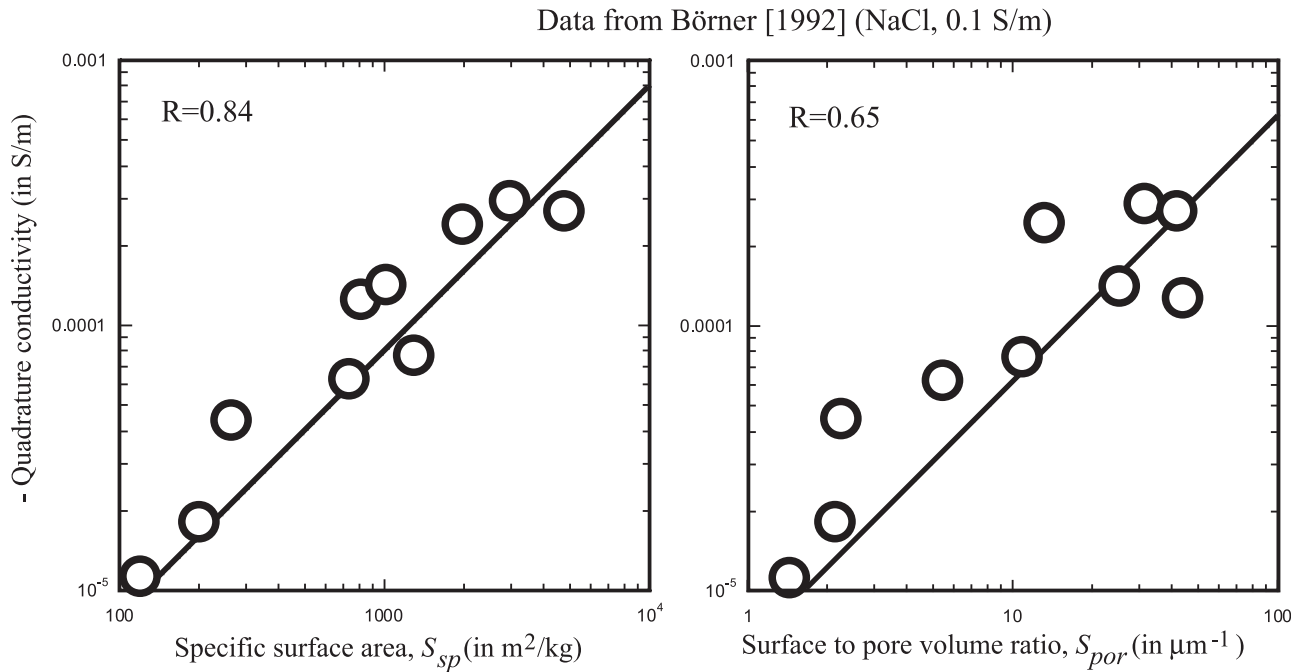


Figure 12. Quadrature conductivity (left) versus specific surface area and (right) versus the surface to pore volume ratio. The solid lines correspond to the linear relationships $\sigma'' \approx -a S_{sp}$ and $\sigma'' \approx -a S_{por}$, respectively, as discussed in the main text. Note that the correlation between the quadrature conductivity is much better than with the surface to pore volume ratio, in agreement with the prediction of the model.

This critical value is independent of the microstructure of the porous material. *Vinegar and Waxman* [1984] analyzed the experimental values obtained by *Fraser* [1964] at 10^{-4} N NaCl to show that this critical value is on the order of -30 mrad. Taking $\beta_{(+)}^S(\text{Na}^+) = 1.5 \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$ and $\beta_{(+)}(\text{Na}^+) = 5.2 \times 10^{-8} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, this yields $f = 0.91$, a value consistent with the estimates given above. The existence of a maximum in the value of the phase lag on the order of -30 mrad is also shown in the values of the phase lag obtained by *Börner* [1992] (see Figure 13). For kaolinite samples, taking $f = 0.98$, the phase lag can reach higher values close to 140 mrad.

3.5. Influence of Temperature

[43] The temperature dependence of the mobility of sodium in the pore water and the apparent surface mobility entering in the surface conductivity term of the in-phase conductivity are given by [*Sen and Goode, 1992; Revil et al., 1998*]

$$\beta_{(+)}(T) = \beta_{(+)}(T_0)[1 + \alpha_f(T - T_0)], \quad (50)$$

$$\beta_S(T) = \beta_S(T_0)[1 + \alpha_S(T - T_0)], \quad (51)$$

with $T_0 = 25^\circ\text{C}$ and $\alpha_f = 0.019/^\circ\text{C}$ [*Sen and Goode, 1992; Binley et al., 2010*] and α_S represent the temperature sensitivity coefficient ($\text{in}/^\circ\text{C}$) for the mobility in the bulk pore water and in the diffuse layer [*Revil et al., 1998*], respectively. As f_M is temperature independent (because it is associated only with two crystalline parameters), it is expected that $\alpha_f \approx \alpha_S$ because the viscosity in the bulk pore water

and in the diffuse layer are the same. At very low salinity, f could be temperature dependent and therefore the two temperature sensitivity coefficients could be different.

[44] The mobility of the counterions in the stern layer is written as

$$\beta_S^{(+)}(T) = \beta_S^{(+)}(T_0)[1 + \alpha_S^{(+)}(T - T_0)], \quad (52)$$

where $\alpha_S^{(+)}$ is unknown. At high salinity, the in-phase conductivity is dominated by the brine conductivity and therefore its temperature dependence is controlled by α_f . Therefore, the temperature dependence of the phase lag at high salinity is given by

$$\varphi(T) = \varphi(T_0) \left[\frac{1 + \alpha_f(T - T_0)}{1 + \alpha_S^{(+)}(T - T_0)} \right]. \quad (53)$$

At low salinity, the in-phase conductivity is dominated by the surface conductivity associated with the diffuse layer and therefore its temperature dependence is controlled by α_S . Therefore, the temperature dependence of the phase lag at high salinity is given by

$$\varphi(T) = \varphi(T_0) \left[\frac{1 + \alpha_S(T - T_0)}{1 + \alpha_S^{(+)}(T - T_0)} \right]. \quad (54)$$

A simple assumption is to consider that $\alpha_f = \alpha_S = \alpha_S^{(+)} = 0.023/^\circ\text{C}$ at high salinities to reduce the number of unknowns (this value is also slightly higher than provided above $\alpha_f = 0.019/^\circ\text{C}$). *Zisser et al.* [2010] found that the

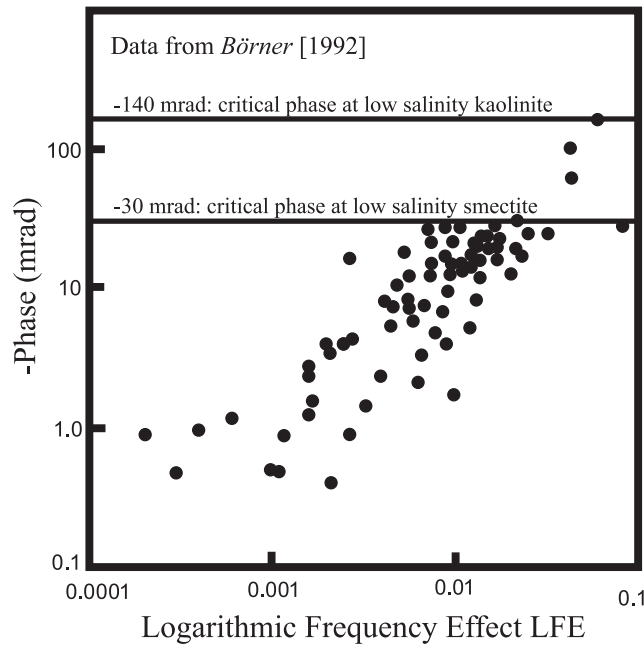


Figure 13. Measured phase lag for the sandstones investigated by Börner [1992]. Note that most of the measurements are below the critical phase lag value predicted by the model developed in this paper (approximately -30 mrad at 25°C for smectite and -140 mrad for kaolinite). The logarithmic frequency effect (LFE) is defined as $\text{LFE}_{12} = [\log |\sigma^*(\omega_2)| - \log |\sigma^*(\omega_1)|] / [\log \omega_2 - \log \omega_1]$.

temperature dependence of the quadrature conductivity of the Bentheimer sandstone (silica: 96%, feldspar: 3%, clay: 1%) is equal to $\alpha_S^{(+)} = 0.0254/^\circ\text{C}$ (in the temperature range 0 – 80°C , at 1 Hz, close to the relaxation peak), therefore close to the temperature dependence of the counterions in the pore water. Binley *et al.* [2010] found for 7 sandstones (Berea, Cottaer, and Sherwood) that the temperature dependence of the in-phase conductivity was $0.022 \pm 0.002/^\circ\text{C}$ and the temperature dependence of the quadrature conductivity was $\alpha_S^{(+)} = 0.023 \pm 0.002/^\circ\text{C}$. For the low-temperature range 0°C to 25°C , Hayley *et al.* [2007] found that $\alpha_f = \alpha_S = 0.020 \pm 0.002/^\circ\text{C}$ for the in-phase conductivity.

[45] With the assumption that $\alpha_f = \alpha_S = \alpha_S^{(+)}$, the phase lag is expected to be nearly temperature independent. This seems to be a good approximation at least for small temperature changes (see Figure 14, which is based on the data reported by Vinegar and Waxman [1984]).

3.6. Prediction of Permeability

[46] The determination of permeability from the complex conductivity is a topic of active research in geophysics [see Börner, 1992; Slater and Lesmes, 2002; Kemna *et al.*, 2005; Binley *et al.*, 2005; Börner, 2006; Hördt *et al.*, 2007; Slater, 2007]. I am only looking here for a simple scaling law in the present paper that is consistent with dimensional analysis (most empirical relationships in the literature do not satisfy dimensional analysis). The permeability scales as the function of the hydraulic radius square (R^2) divided by the formation factor F (Kozeny-Carman equation [e.g., Bernabé and Revil, 1995]) Using the scaling

of the hydraulic radius with the charge density Q_V ($R \propto 1/Q_V$), the scaling of the quadrature conductivity with the charge density Q_V , and the scaling of the formation factor with the porosity (using a high-porosity asymptotic behavior), I propose the following scaling law between the permeability, the formation factor, and the quadrature conductivity:

$$k = \frac{1}{a} \frac{1}{F^3 (\sigma'')^2}, \quad (55)$$

where a is a universal constant. If I consider σ'' expressed in S m^{-1} , the permeability expressed in mD (approximately 10^{-15} m^2) then using $a = 120 \text{ S}^2 \text{ m}^{-2} \text{ mD}^{-1}$ allows to fit the permeability data of Vinegar and Waxman [1984] reported in Table 1 and those of Börner [1992] reported in Table 2 (see Figure 15). The predictive power of equation (55) is inside an order of magnitude and no more accuracy can be probably expected. Equation (55) is different from the equation proposed by Slater and Lesmes [2002] (the permeability appears as a power law function of only the quadrature conductivity) but is consistent with equation (79) of Revil and Florsch [2010].

3.7. Influence of Saturation

[47] I first assume that the presence of a second nonwetting phase does not change the cation exchange capacity or the surface area able to influence the quadrature conductivity. With this additional assumption, the influence of saturation is easy to take into account using the following transform in the expressions of the in-phase and quadrature conductivity [Vinegar and Waxman, 1984; Revil, 1999]:

$$\frac{1}{F} \Rightarrow \frac{1}{F} S_w^n, \quad (56)$$

$$Q_V \Rightarrow Q_V S_w^{-1}, \quad (57)$$

where S_w is the relative water saturation ($S_w = 1$ at full water saturation, $0 \leq S_w \leq 1$). Equation (57) is called the second Archie's law and n is called the saturation or second Archie exponent (in absence of information it is customary to assume that $n \approx m \approx 2$ [e.g., Waxman and Smits, 1968]). Because the term $[\phi/(1-\phi)]$, at high porosity is proportional to $1/F$, I apply the transform given by equation (56) to this term. Using equations (27), (56), and (57) yields

$$\sigma''(S_w) \approx -\frac{2}{3} \left(\frac{\phi}{1-\phi} \right) S_w^{n-1} \lambda Q_V, \quad (58)$$

$$\sigma''(S_w) \approx S_w^p \sigma''_{\text{Sat}}, \quad (59)$$

with $p = n - 1$, in agreement with the results of Vinegar and Waxman [1984]. We test in Table 3 the prediction of equation (55) with $p = n - 1$ for some sands and sandstones. The prediction of this simple model is shown in Figure 16 using the data reported in Table 3. Equation (59) seems in agreement with not only the results of Vinegar and Waxman [1984] but also those of Slater and Glaser [2003] and Schmutz *et al.* [2010]. That said, the previous

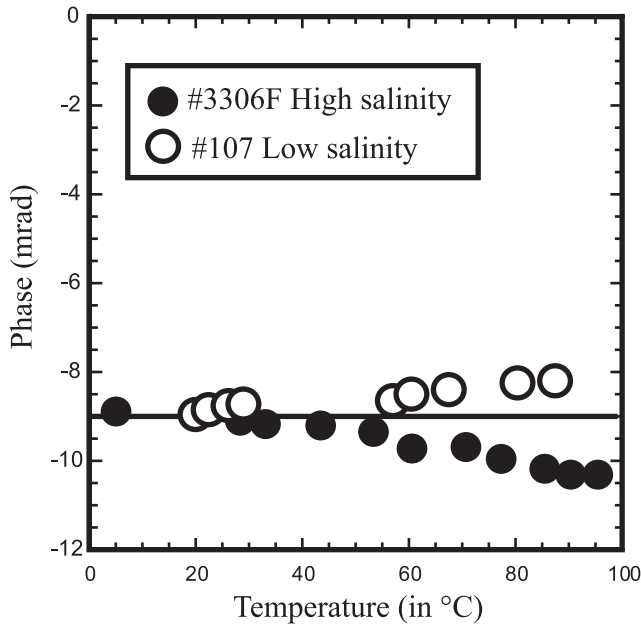


Figure 14. Temperature dependence of the phase lag for two shaly sandstones investigated by *Vinegar and Waxman* [1984]. At first approximation and for small temperature changes, the phase lag is temperature independent (line). This means in turn that the temperature dependence of the quadrature conductivity is roughly the same as the temperature dependence of the in-phase conductivity. The size of the symbols is on the order of the magnitude of the uncertainty.

model equation does not include the effect of the CEC attached to the interface between the wetting and the non-wetting phases and a more complete model needs to be developed to account for this contribution.

3.8. Tests of POLARIS

[48] For isothermal conditions, I test now the ability of POLARIS to predict the quadrature conductivity measurements reported by *Vinegar and Waxman* [1984]. The input parameters are the excess charge density Q_V , the salinity C_f , and the porosity ϕ . Nothing else is required. Indeed, combining equations (28) and (39) yields

$$\sigma'' \approx -\frac{2}{3} \lambda_M \left(\frac{\phi}{1-\phi} \right) \left(\frac{K_{Na} C_f}{10^{-\text{pH}}(1-f_M) + K_H + K_{Na} C_f} \right) Q_V. \quad (60)$$

Equation (60) provides an explicit dependence of the quadrature conductivity with the porosity, the salinity, and excess charge density for pH values close to neutral (5 to 8). At 25°C, to perform my comparison I use the model parameters determined in section 3.1, $\lambda_M = (1.32 \pm 0.03) \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $K_{Na} = (3.0 \pm 0.4) \times 10^{-4}$, $K_H = 8 \times 10^{-6}$, and pH = 5.6 (pH of water in equilibrium with the atmosphere, slightly acidic because of the solubility of the CO_2 in the pore water). The comparison between the predicted quadrature conductivity and the measured quadrature conductivity is shown in Figure 17. The model offers

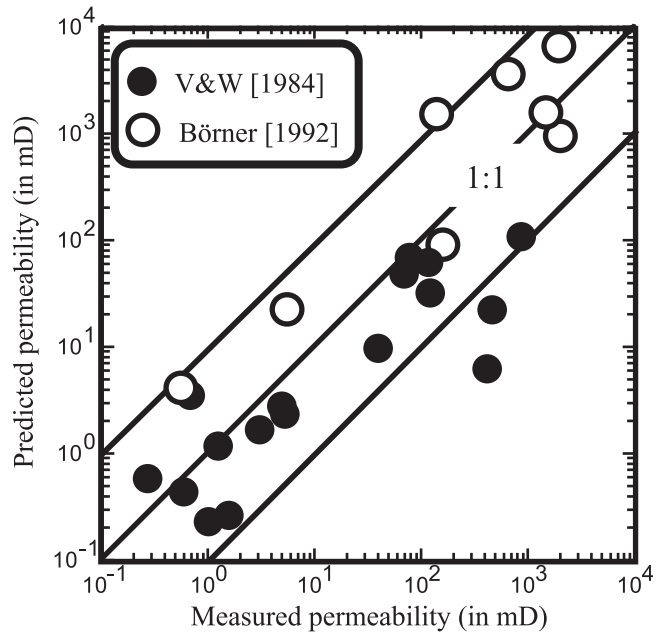


Figure 15. Predicted permeability (using the formation factor and the quadrature conductivity) versus the measured brine permeability (in mD, $1 \text{ mD} \approx 10^{-12} \text{ m}^2$) using the databases of *Vinegar and Waxman* [1984] and *Börner* [1992]. The data covers 5 orders of magnitude in permeability. The confidence interval corresponds to 1 order of magnitude ($R = 0.69$).

an excellent prediction of the data with very simple input parameters.

[49] Equation (60) is also tested with the data shown in Figure 4 to predict the salinity dependence of the quadrature conductivity. I used the measured values of the porosity and excess charge density Q_V and $\lambda_M = (1.32 \pm 0.03) \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $K_{Na} = (3.0 \pm 0.4) \times 10^{-4}$, $K_H = 8 \times 10^{-6}$, and pH = 5.6 for samples 3847A and B2. The model prediction works quite well for both samples.

[50] Equation (60) can be also written directly as a function of the CEC without using the porosity as an input parameter. This means that the CEC and the salinity are truly the two fundamental parameters needed to compute the quadrature conductivity, all the other parameters (pH, λ_M , K_{Na} , K_H) are known as discussed above. The quadrature conductivity can be also computed directly from the specific surface area. Using equations (46) and (47) in equation (60) yields

$$\sigma'' \approx -\frac{2}{3} \lambda_M \left(\frac{K_{Na} C_f}{10^{-\text{pH}} + K_{Na} C_f} \right) Q_S \rho_g S_{sp}. \quad (61)$$

In Figure 17 (right), I use $\lambda_M = (1.32 \pm 0.03) \times 10^{-10} \text{ m}^2 \text{ s}^{-1} \text{ V}^{-1}$, $K_{Na} = (3.0 \pm 0.4) \times 10^{-4}$, $K_H = 8 \times 10^{-6}$, pH = 5.6 (exactly like in the case above), $\rho_g = 2650 \text{ kg m}^{-3}$, and $Q_S = 0.8 \text{ C m}^2$ (5 charges nm^{-2}) to fit the data of *Börner* [1992]. The only input parameters are the salinity and the specific surface area, both independently measured by *Börner* [1992]. Again, there is a good agreement between the predictions of POLARIS and the experimental data.

Table 3. Prediction of the Exponent $p = n - 1$ for the Quadrature Conductivity as a Function of the Saturation Exponent n in the In-Phase Conductivity and Comparison With the Measured Value Using Quadrature Conductivity Data at Different Saturations of the Pore Water

Sample	Type ^a	Nonwetting Fluid	Measured n	Predicted p	Measured p
N2 55–60	sand (1)	air	1.3 ± 0.1	0.3 ± 0.1	0.6 ± 0.1
A and B	sand (2)	oil	2.1 ± 0.1	1.1 ± 0.1	1.3 ± 0.1
V104	clayey sand (3)	crude oil	2.17 ± 0.05	1.17 ± 0.05	1.18 ± 0.05
V106	clayey sand (3)	crude oil	2.43 ± 0.05	1.43 ± 0.05	1.28 ± 0.05
V110	clayey sand (3)	Soltrol	2.11 ± 0.05	1.11 ± 0.05	1.27 ± 0.05
V112	clayey sand (3)	Soltrol	1.63 ± 0.05	0.63 ± 0.05	0.78 ± 0.05
1883	clayey sand (3)	Soltrol	1.81 ± 0.05	0.81 ± 0.05	1.20 ± 0.05
3371	clayey sand (3)	Soltrol	2.40 ± 0.05	1.40 ± 0.05	1.17 ± 0.05
3372C	clayey sand (3)	Soltrol	2.35 ± 0.05	1.35 ± 0.05	1.16 ± 0.05
3374	clayey sand (3)	Soltrol	1.79 ± 0.05	0.79 ± 0.05	0.93 ± 0.05
3307C	clayey sand (3)	Soltrol	1.89 ± 0.05	0.89 ± 0.05	1.03 ± 0.05
5P-3A	sand (4)	heavy oil	1.67 ± 0.07	0.67 ± 0.07	0.59 ± 0.05
P-I-A	sand (4)	heavy oil	1.67 ± 0.07	0.67 ± 0.07	0.59 ± 0.05
Sample 1	sand (5)	air	1.38 ± 0.02	0.38 ± 0.02	0.61 ± 0.01

^aThe numbers in parentheses indicate the following: 1, unconsolidated sediments from the Olathe study site [from Slater and Glaser, 2003]; 2, unconsolidated sand [from Schmutz et al., 2010]; (3) clayey sands [from Vinegar and Waxman, 1984]; 4, tar sands (from Martinez, F. J., M. L. Batzle, and A. Revil, Influence of temperature on seismic velocities and complex conductivity of heavy oil bearing sands, submitted to *Geophysics*, 2012); 5, sand [Abdel-Aal et al., 2006].

4. Discussion

[51] A simple induced polarization model named POLARIS has been introduced to explain the dependence of the complex, frequency-independent, conductivity with the microstructure (porosity, specific surface area), interfacial properties (charge density or CEC) as well as with environmental variables (temperature and salinity). This is the first time that such a model is achieved keeping both scientific rigor and simplicity and it is therefore expected to be broadly applied, as discussed further below, to various problems.

[52] I discuss briefly two potential applications of POLARIS. This model can be used for mapping soil properties in agriculture (like their CEC because of the direct relationship between the quadrature conductivity and the CEC). Indeed, because of the broad range of polarization length scales for clayey soils, it is expected that the phase and the

quadrature conductivity are rather frequency independent. Mapping the phase lag can be done therefore at 10 Hz for instance. It is however possible that, in some circumstances, the Maxwell Wagner polarization [see Chen and Or, 2006] will have to be accounted for to interpret the phase lag measurements at this frequency. Additional information regarding the water content can be obtained from GPR for instance. In the case of clay-free soil, the mobility of the counterions in the Stern layer is expected to be the same that the mobility of the same ions in the pore water (like for silica). This would explain the high polarization observed in this type of soils (L. Slater, personal communication, 2011).

[53] The model described above can be applied to monitor salt tracer tests. Recently, Karaoulis et al. [2011] have developed a time-lapse spectral induced polarization inversion algorithm to monitor and image changes in the induced polarization parameters over time. This inversion algorithm can be combined with the model developed above to follow salt tracer tests at a given frequency (e.g., 1 Hz). This approach would be much more powerful than the traditional approach which is to use DC resistance tomography usually neglecting surface conductivity. A very innovative approach would be to use time-lapse inversion of both induced polarization and self-potential data because such geophysical measurements can be performed with the same apparatus and the physics of self-potential signals to follow salt tracer tests is now well understood [see Revil and Jardani, 2010].

5. Concluding Statements

[54] I have developed an analytical model called POLARIS for the quadrature conductivity on the basis of the solution of the differential effective medium theory. This model accounts for the two components of the electrical double layer coating the pore water mineral interface. The following conclusions have been reached.

[55] 1. The mobility of the counterions in the Stern layer is much lower (by a factor 350 for sodium) than the

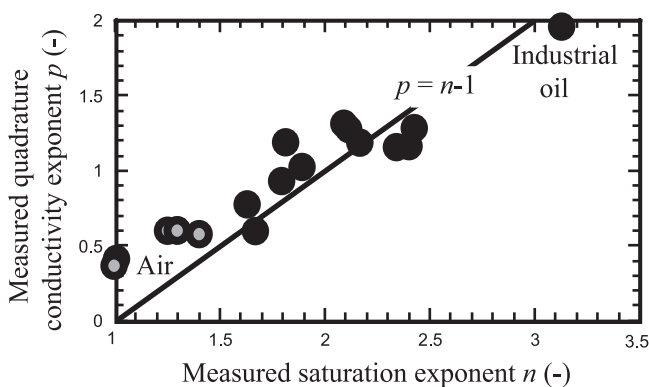


Figure 16. Predicted versus measured saturation exponent p for the quadrature conductivity. The predicted value is determined using the relationship $p = n - 1$, where n is the saturation exponent determined from the in-phase conductivity dependence on saturation. The uncertainty is roughly equal to the size of the symbols, and $R = 0.81$. References can be found in Table 3.

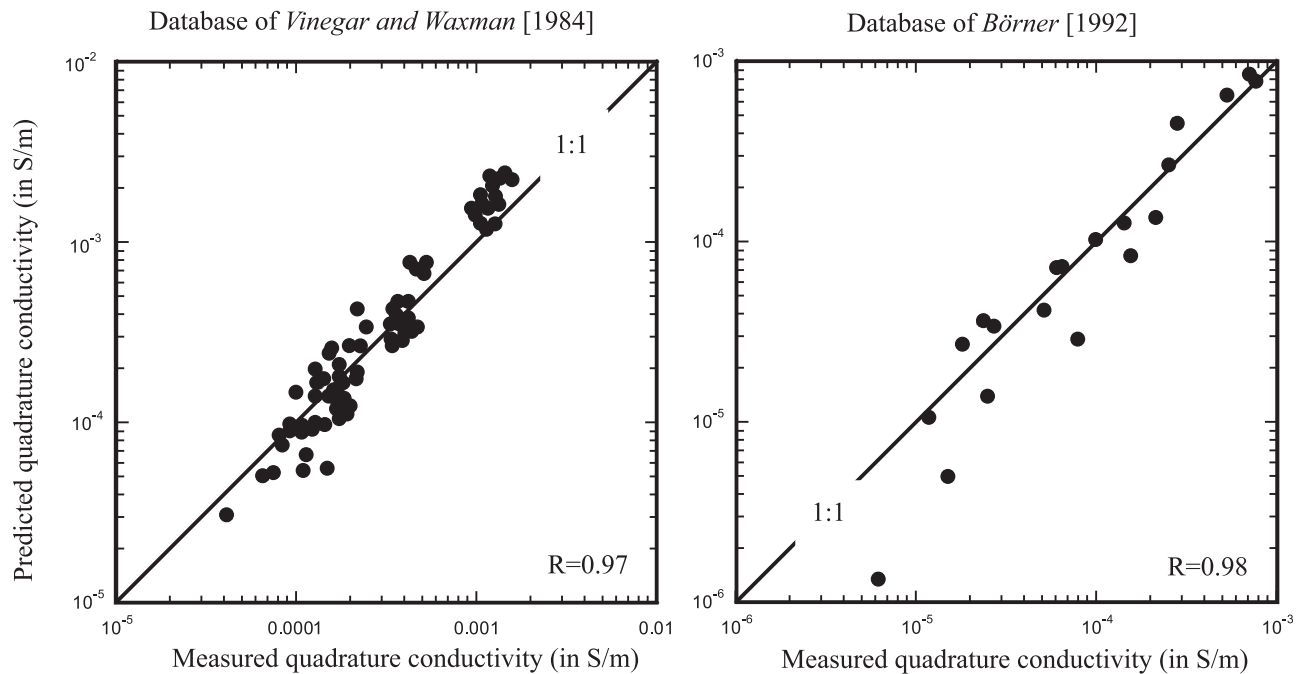


Figure 17. Predicted versus measured quadrature conductivity using, as input variables, the porosity, the excess charge density, and the salinity of the pore water for the quadrature conductivity data from Vinegar and Waxman [1984] and the specific surface area and the salinity for the data from Börner [1992] (samples E10, E12, and B2).

mobility of the counterions in the pore water in a clay material. This is in sharp contrast with the case of silica for which the mobility of the counterions in the Stern layer is the same than in the bulk pore water. This observations is explained by the fraction of counterions in the Stern layer, which is much more important in clays by comparison with silica grains.

[56] 2. The surface conductivity entering into the in-phase component of the electrical conductivity is controlled by the diffuse layer (the contribution of the Stern layer can be safely neglected because of the very small mobility of the counterions in the Stern layer).

[57] 3. The polarization of the Stern layer represents the dominating polarization mechanism in shaly sands except maybe at very high salinities for which membrane polarization is expected to dominate the overall polarization response. The quadrature conductivity seems to be controlled by (1) the mobility of the counterions in the Stern layer, (2) the excess of charge per unit volume determined by the CEC, (3) the porosity, and (4) by the fraction of counterions present in the Stern layer (partition coefficient).

[58] 4. The partition coefficient can be determined from a simple sorption isotherm, which predicts its salinity dependence and therefore the salinity dependence of the quadrature conductivity. From this model, it is possible to predict the effect of the composition of the pore water upon the quadrature conductivity using a sorption isotherm that would take into account the composition of the pore water. This step will be investigated in a future contribution.

[59] 5. The model explains the small dependence of the phase lag with the temperature and the observed dependence of the quadrature conductivity with the specific surface area of the material.

[60] 6. Permeability can be predicted inside 1 order of magnitude using the formation factor (determined from the in phase conductivity) and the quadrature conductivity.

[61] The model appears very robust in being able to predict the dependency of the complex conductivity with both textural, electrochemical, and environmental parameters. The next steps will be (1) to develop a model for the membrane polarization and to combine this model with the Stern layer polarization model developed above to get a unified model of low-frequency induced polarization in shaly sands, clean sands, and clay-sand mixtures and (2) to apply POLARIS to monitor bacteria growth in porous media [see Atekwana and Slater, 2009].

Notation

E	electrical field.
E_0	constant electrical field.
J	conduction current density.
J_d	displacement current density.
H	magnetic field.
$\beta_{(+)}^S$	mobility of the counterions in the Stern layer.
$\beta_{(+)}$	mobility of the cation in the pore water.
β_S	apparent surface mobility for the double layer.
ϵ^*	complex permittivity.
ϵ'	real part of the permittivity.
ϵ''	imaginary part of the permittivity.
ϵ_0	permittivity of vacuum.
ϵ_{eff}	apparent permittivity.
ϕ	connected porosity.
λ	apparent mobility for the counterions.
λ_M	maximum value reached by the apparent mobility.
σ^*	complex electrical conductivity.

σ'	real part of the conductivity.
σ''	imaginary part of the conductivity.
$ \sigma $	amplitude of the complex conductivity.
σ_{eff}	apparent (real) conductivity.
σ_f	pore water (real) conductivity.
σ_s^*	surface (complex) conductivity.
σ'_S	real (in phase) surface conductivity.
σ''_S	imaginary (quadrature) surface conductivity.
ω	angular frequency.
φ	phase lag.
ρ_g	the grain density.
Σ_S	specific surface conductivity of the Stern layer.
Γ_S^0	total surface site density of the mineral surface.
Γ_{SNa}^0	surface charge density of the sites $>S^-Na^+$.
Γ_{S-}^0	surface charge density of the sites $>S^-Na^+$.
Γ_{SH}^0	surface charge density of the sites $>SH^0$.
Γ_X^0	equivalent surface charge density for the isomorphic substitutions.
α_S	temperature sensitivity coefficient for the mobility of the counterions.
α_f	temperature sensitivity coefficient for the ionic mobilities in the pore water.
α_S	temperature sensitivity coefficient for the mobility of the counterions in the Stern layer.
Du	Dukhin number.
d_0	single grain diameter.
D	grain diameter.
C_f	salinity.
E_h	expectation of $f(D)/D$.
F	formation factor.
f	partition coefficient of the electrical double layer.
f_M	maximum value reached by the partition coefficient.
$f(D)$	grain diameter probability distribution.
i	pure imaginary number.
k	permeability.
K_{Na}	equilibrium sorption coefficient for Na^+ .
K_H	equilibrium sorption coefficient for H^+ .
m	cementation exponent.
n	saturation exponent.
p	quadrature conductivity saturation exponent.
pH	Log of the activity of the proton concentration.
Q_V	total volumetric (Stern and diffuse layers) charge density.
$\bar{Q}_V$	volumetric diffuse layer charge density.
Q_S	charge density per unit surface area.
R	hydraulic radius.
S_w	relative water saturation.
S_{Sp}	specific surface area of the material.
S_{por}	specific surface per unit pore volume.
t	time.
T	temperature.
T_0	reference temperature.

[62] **Acknowledgments.** I thank the Office of Science (BER), US Department of Energy (awards DE-FG02-08ER646559 and DE-SC0007118), and Chevron Energy Technology Company for financial support. I thank the Associate Editor, Fred Day-Lewis, and the referees (Lee Slater, Andrew Binley, and an anonymous referee) for their very constructive reviews.

References

- Abdel Aal, G. Z., L. D. Slater, and E. A. Atekwana (2006), Induced-polarization measurements on unconsolidated sediments from a site of active hydrocarbon biodegradation, *Geophysics*, *71*, H13–H24, doi:10.1190/1.2187760.
- Archie, G. E. (1942), The electrical resistivity log as an aid in determining some reservoir characteristics, *Trans. Am. Inst. Min. Metall. Eng.*, *146*, 54–62.
- Atekwana, A., and L. Slater (2009), Biogeophysics: A new frontier in earth science research, *Rev. Geophys.*, *47*, RG4004, doi:10.1029/2009RG000285.
- Bernabé, Y., and A. Revil (1995), Pore-scale heterogeneity, energy dissipation and the transport properties of rocks, *Geophys. Res. Lett.*, *22*(12), 1529–1552.
- Binley, A., L. D. Slater, M. Fukes, and G. Cassiani (2005), Relationship between spectral induced polarization and hydraulic properties of saturated and unsaturated sandstone, *Water Resour. Res.*, *41*, W12417, doi:10.1029/2005WR004202.
- Binley, A., S. Kruschwitz, D. Lesmes, and N. Kettridge (2010), Exploiting the temperature effects on low frequency electrical spectra of sandstone: A comparison of effective diffusion path lengths, *Geophysics*, *75*, A43–A46, doi:10.1190/1.3483815.
- Bolève, A., A. Crespy, A. Revil, F. Janod, and J. L. Mattiuzzo (2007), Streaming potentials of granular media: Influence of the Dukhin and Reynolds numbers, *J. Geophys. Res.*, *112*, B08204, doi:10.1029/2006JB004673.
- Börner, F. D. (1992), Complex conductivity measurements of reservoir properties, in *Proceedings of the Third European Core Analysis Symposium*, pp. 359–386, Advances in Core Evaluation v. 3: Reservoir Management—Reviewed Proceedings of the Society for Core Analysis Third European Core Analysis Symposium—Paris, edited by P. F. Worthington and Catherine Chardaire-Riviere, Harwood Academic, Neward, New Jersey.
- Börner, F. D. (2006), Complex conductivity measurements, in: *Groundwater Geophysics—A Tool for Hydrogeology*, edited by R. Kirsch, 119–153, Springer, Berlin.
- Bourg, I. C., and G. Sposito (2011), Molecular dynamics simulations of the electrical double layer on smectite surfaces contacting concentrated mixed electrolyte ($NaCl-CaCl_2$) solutions, *J. Colloid Interface Sci.*, *360*, 701–715.
- Brovelli, A., and G. Cassiani (2011), Combined estimation of effective electrical conductivity and permittivity for soil monitoring, *Water Resour. Res.*, *47*, W08510, doi:10.1029/2011WR010487.
- Bruggeman, D. A. G. (1935), Berechnung verschiedener physikalischer konstanten von heterogenen substansen, *Ann. Phys. (Leipzig)*, *24*, 636–679.
- Bussian, A. E. (1983), Electrical conductance in a porous media, *Geophysics*, *48*, 1258–1268.
- Chen, Y., and D. Or (2006), Effects of Maxwell-Wagner polarization on soil complex dielectric permittivity under variable temperature and electrical conductivity, *Water Resour. Res.*, *42*, W06424, doi:10.1029/2005WR004590.
- de Lima, O. A. L., and M. M. Sharma (1990), A grain conductivity approach to shaly sandstone, *Geophysics*, *55*, 1347–1356.
- de Lima, O. A. L., and M. M. Sharma (1992), A generalized Maxwell-Wagner theory for membrane polarization in shaly sands, *Geophysics*, *57*, 431–440.
- Drolon, H., B. Hoyez, F. Druaux, and A. Faure (2003), Multiscale roughness analysis of particles: Application to the classification of detrital sediments, *Math. Geol.*, *35*(7), 805–817.
- Dukhin, S. S., and V. N. Shilov (2002), Nonequilibrium electric surface phenomena and extended electrokinetic characterization of particles, in *Interfacial Electrokinetics and Electrophoresis*, Surfactant Sci. Ser., vol. 106, edited by A. V. Delgado, pp. 55–85, Marcel Dekker, New York.
- Fraser, D. C. (1964), Electrical properties of clay-containing sandstone: B. Electrical polarization, *Rep. MT-64-4*, Inst. of Eng. Res., Univ. of Calif., Berkeley.
- Garrouch, A. A., and M. M. Sharma (1994), The influence of clay content, salinity, stress and wettability on the dielectric properties of brine-saturated rocks: 10 Hz to 10 MHz, *Geophysics*, *59*, 909–917.
- Glover, P. W. J., M. J. Hole, and J. Pous (2000), A modified Archie's law for two conducting phases, *Earth Planet. Sci. Lett.*, *180*, 369–384.
- Grosse, C. (2011), Extension of the classic thin double layer polarization theory of colloidal suspensions to include the stagnant layer conductivity, *J. Phys. Chem. B.*, *115*, 8996–9004.

- Hanai, T. (1968), Electrical properties of emulsions, in *Emulsion Science*, edited by P. Sherman, pp. 354–478, Academic, London.
- Hayley, K., L. R. Bentley, M. Gharihi, and M. Nightingale (2007), Low temperature dependence of electrical resistivity: Implications for near surface geophysical monitoring, *Geophys. Res. Lett.*, **34**, L18402, doi:10.1029/2007GL031124.
- Hördt, A., R. Blaschek, A. Kemna, and N. Zisser (2007), Hydraulic conductivity estimation from induced polarisation data at the field scale—The Krauthausen case history, *J. Appl. Geophys.*, **62**, 33–46.
- Hyslip, J. P., and L. E. Vallejo (1997), Fractal analysis of the roughness and size distribution of granular materials, *Eng. Geol.*, **48**(3–4), 231–244.
- Jougnot, D., A. Ghorbani, A. Revil, P. Leroy, and P. Cosenza (2010), Spectral induced polarization of partially saturated clay-rocks: A mechanistic approach, *Geophys. J. Int.*, **180**, 210–224, doi:10.1111/j.1365-246X.2009.04426.x.
- Kan, R., and P. N. Sen (1987), Electrolytic conduction in periodic arrays of insulators with charges, *J. Chem. Phys.*, **86**, 5748–5756.
- Karaoulis, M., A. Revil, D. D. Werkema, B. Minsley, W. F. Woodruff, and A. Kemna (2011), Time-lapse 3D inversion of complex conductivity data using an active time constrained (ATC) approach, *Geophys. J. Int.*, **187**, 237–251, doi:10.1111/j.1365-246X.2011.05156.x.
- Kemna, A. (2000), Tomographic inversion of complex resistivity: theory and application, PhD thesis, 196 pp., Ruhr-Univ. Bochum, Bochum, Germany.
- Kemna, A., H.-M. Münch, K. Titov, E. Zimmermann, and H. Vereecken (2005), Relation of SIP relaxation time of sands to salinity, grain size and hydraulic conductivity, in *Proceedings of the 11th European Meeting of Environmental and Engineering Geophysics, Extended Abstracts: Near Surface 2005: P054*, 11th European Meeting of Environmental and Engineering Geophysics, Palermo, Italy.
- Le Méhauté, A., and G. Crépy (1983), Introduction to transfer and motion in fractal media: The geometry of kinetics, *Solid State Ionics*, **9**–10, 17–30.
- Leroy, P., and A. Revil (2004), A triple layer model of the surface electrochemical properties of clay minerals, *J. Colloid Interface Sci.*, **270**(2), 371–380.
- Leroy, P., and A. Revil (2009), Spectral induced polarization of clays and clay-rocks, *J. Geophys. Res.*, **114**, B10202, doi:10.1029/2008JB006114.
- Leroy, P., A. Revil, S. Altmann, and C. Tournassat (2007), Modeling the composition of the pore water in a clay-rock geological formation (Callovo-Oxfordian, France), *Geochim. Cosmochim. Acta*, **71**(5), 1087–1097, doi:10.1016/j.gca.2006.11.009.
- Leroy, P., A. Revil, A. Kemna, P. Cosenza, and A. Gorbani (2008), Spectral induced polarization of water-saturated packs of glass beads, *J. Colloid Interface Sci.*, **321**(1), 103–117.
- Lesmes, D. P., and F. D. Morgan (2001), Dielectric spectroscopy of sedimentary rocks, *J. Geophys. Res.*, **106**(B7), 13,329–13,346.
- Marshall, D. J., and T. R. Madden (1959), Induced polarization, a study of its causes, *Geophysics*, **24**, 790–816.
- Meier, L. P., and G. Kahr (1999), Determination of the cation exchange capacity (CEC) of clay minerals using the complexes of copper(II) ion with triethylenetetramine and tetraethylenepentamine, *Clays Clay Miner.*, **47**(3), 386–388.
- Morgan, F. D., F. Scira-Scappuzzo, W. Shi, W. Rodi, J. Sogade, Y. Vichabian, and D. Lesmes (1999), Induced polarization imaging of a jet fuel plume, Symposium on the Application of Geophysics to Engineering and Environmental Problems “SAGEEP,” Environmental and Engineering Geophysical Society, Proceedings, 541–548, Oakland, California.
- Niwas, N., P. K. Gupta, and O. A. L. de Lima (2007), Nonlinear electrical conductivity response of shaly-sand reservoir, *Curr. Sci.*, **92**(5), 612–617.
- Olhoeft, G. R. (1979), Electrical properties, in *Initial Report of the Petrophysics Laboratory*, U.S. Geol. Surv. Circ., 789, 1–26.
- Olhoeft, G. R. (1986), Direct detection of hydrocarbon and organic chemicals with ground-penetrating radar and complex resistivity: Petroleum, hydrocarbons and organic chemicals, in *Proceedings of the NWWA/API Conference on Petroleum Hydrocarbons and Organic Chemicals in Ground Water—Prevention, Detection, and Restoration*, pp. 284–305, Natl. Well Water Assoc., Dublin, Ohio.
- Pelton, W. H., S. H. Ward, P. G. Hallor, W. R. Sill, and P. H. Nelson (1978), Mineral discrimination and removal of inductive coupling with multifrequency IP, *Geophysics*, **43**, 588–609.
- Revil, A. (1999), Ionic diffusivity, electrical conductivity, membrane and thermoelectric potentials in colloids and granular porous media: A unified model, *J. Colloid Interface Sci.*, **212**, 503–522.
- Revil, A. (2001), Pervasive pressure solution transfer in a quartz sand, *J. Geophys. Res.*, **106**(B5), 8665–8690.
- Revil, A., and L. M. Cathles (1999), Permeability of shaly sands, *Water Resour. Res.*, **35**(3), 651–662.
- Revil, A., D. Hermitte, E. Spangenberg, and J. J. Cochémé (2002), Electrical properties of zeolitized volcanoclastic materials, *J. Geophys. Res.*, **107**(B8), 2168, doi:10.1029/2001JB000599.
- Revil, A., and P. Leroy (2004), Governing equations for ionic transport in porous shales, *J. Geophys. Res.*, **109**, B03208, doi:10.1029/2003JB002755.
- Revil, A., and N. Florsch (2010), Determination of permeability from spectral induced polarization in granular media, *Geophys. J. Int.*, **181**, 1480–1498, doi:10.1111/j.1365-246X.2010.04573.x.
- Revil, A., and A. Jardani (2010), Stochastic inversion of permeability and dispersivities from time lapse self-potential measurements: A controlled sandbox study, *Geophys. Res. Lett.*, **37**, L11404, doi:10.1029/2010GL043257.
- Revil, A., and M. Skold (2011), Salinity dependence of spectral induced polarization in sands and sandstones, *Geophys. J. Int.*, **187**, 813–824, doi:10.1111/j.1365-246X.2011.05181.x.
- Revil, A., L. M. Cathles, S. Losh, and J. A. Nunn (1998), Electrical conductivity in shaly sands with geophysical applications, *J. Geophys. Res.*, **103**(B10), 23,925–23,936.
- Revil, A., W. F. Woodruff, and N. Lu (2011), Constitutive equations for coupled flows in clay materials, *Water Resour. Res.*, **47**, W05548, doi:10.1029/2010WR010002.
- Schmutz, M., A. Revil, P. Vaudelet, M. Batzle, P. Femenía Viñao, and D. D. Werkema (2010), Influence of oil saturation upon spectral induced polarization of oil bearing sands, *Geophys. J. Int.*, **183**, 211–224, doi:10.1111/j.1365-246X.2010.04751.x.
- Schwarz, G. (1962), A theory of the low-frequency dielectric dispersion of colloidal particles in electrolyte solution, *J. Phys. Chem.*, **66**, 2636–2642.
- Scott, J., and R. Barker (2003), Determining pore-throat size in Permian-Triassic sandstones from low-frequency electrical spectroscopy, *Geophys. Res. Lett.*, **30**(9), 1450, doi:10.1029/2003GL016951.
- Sen, P. N., and P. A. Goode (1992), Influence of temperature on electrical conductivity on shaly sands, *Geophysics*, **57**, 89–96.
- Sen, P. N., C. Scala, and M. H. Cohen (1981), A self-similar model for sedimentary rocks with application to the dielectric constant of fused glass beads, *Geophysics*, **46**, 781–795.
- Sistat, P., A. Kozmaib, N. Pismenskayab, C. Larchet, G. Pourcellya, and V. Nikonenkob (2008), Low-frequency impedance of an ion-exchange membrane system, *Electrochim. Acta*, **53**, 6380–6390.
- Skold, M., and A. Revil (2011), The pH dependence of spectral induced polarization of silica sands: Experiment and modeling, *Geophys. Res. Lett.*, **38**, L12304, doi:10.1029/2011GL047748.
- Slater, L. (2007), Near surface electrical characterization of hydraulic conductivity: From petrophysical properties to aquifer geometries: A review, *Surv. Geophys.*, **28**(2–3), 169–197.
- Slater, L., and D. Glaser (2003), Controls on induced polarization in sandy unconsolidated sediments and application to aquifer characterization, *Geophysics*, **68**, 1547–1558.
- Slater, L., and D. P. Lesmes (2002), Electrical-hydraulic relationships observed for unconsolidated sediments, *Water Resour. Res.*, **38**(10), 1213, doi:10.1029/2001WR001075.
- Tarasov, A., and K. Titov (2007), Relaxation time distribution from time domain induced polarization measurements, *Geophys. J. Int.*, **170**, 31–43.
- Tertre, E., E. Ferrage, I. Bihannic, L. J. Michot, and D. Prêt (2011), Influence of the ionic strength and solid/solution ratio on Ca(II)-for-Na⁺ exchange on montmorillonite. Part 2: Understanding the effect of the m/V ratio. Implications for pore water composition and element transport in natural media, *J. Colloid Interface Sci.*, **363**, 334–347.
- Thomas, E. C. (1976), Determination of Qv from membrane potential measurements on shaly sands, *Trans. Am. Inst. Min. Metall. Pet. Eng.*, **261**, 1087–1096.
- Titov, K., V. Komarov, V. Tarasov, and A. Levitski (2002), Theoretical and experimental study of time domain induced polarization in water-saturated sands, *J. Appl. Geophys.*, **50**, 417–433.
- Tournassat, C., Y. Chapron, P. Leroy, M. Bizi, and F. Boulahya (2009), Comparison of molecular dynamics simulations with triple layer and modified Gouy-Chapman models in 0.1 M NaCl-montmorillonite system, *J. Colloid Interface Sci.*, **339**(2), 533–541.
- Urtinov, M. A.-K., E. V. Kirillova, N. M. Seidova, and V. V. Nikonenko (2007), Decoupling of the Nernst-Planck and Poisson equations. Application to a membrane system at overlimiting currents, *J. Phys. Chem. B*, **111**, 14,208–14,222.

- Vanhala, H. (1997), Mapping oil-contaminated sand and till with the spectral induced polarization (SIP) method, *Geophys. Prospect.*, *45*, 303–326.
- Vaudelet, P., A. Revil, M. Schmutz, M. Franceschi, and P. Bégassat (2011a), Induced polarization signature of the presence of copper in saturated sands, *Water Resour. Res.*, *47*, W02526, doi:10.1029/2010WR009310.
- Vaudelet, P., A. Revil, M. Schmutz, M. Franceschi, and P. Bégassat (2011b), Changes in induced polarization associated with the sorption of sodium, lead, and zinc on silica sands, *J. Colloid Interface Sci.*, *360*, 739–752.
- Vinegar, H. J., and M. H. Waxman (1984), Induced polarization of shaly sands, *Geophysics*, *49*, 1267–1287.
- Wang, M., J. Liu, and S. Chen (2007), Electrical potential distribution in nanoscale electroosmosis: from molecules to continuum, *Molecular Simulation*, *33*(15), 1273–1277.
- Waxman, M. H., and L. J. M. Smits (1968), Electrical conductivities in oil-bearing shaly sands, *J. Soc. Pet. Eng.*, *8*, 107–122.
- Weller, A., L. Slater, S. Nordsiek, and D. Ntarlagiannis (2010), On the estimation of specific surface per unit pore volume from induced polarization: A robust empirical relation fits multiple data sets, *Geophysics*, *75*, WA105–WA112.
- Wong, P. Z. (1987), Fractal surface in porous media, in *Physics and Chemistry of Porous Media*, vol. 2, edited by J. R. Banavar, J. Koplik, and K. W. Winkler, *AIP Conf. Proc.*, *154*, 304–318.
- Woodruff, W. F., and A. Revil (2011), CEC-normalized clay-water sorption isotherm, *Water Resour. Res.*, *47*, W11502, doi:10.1029/2011WR010919.
- Zisser, N., A. Kemna, and G. Nover (2010), Dependence of spectral-induced polarization response of sandstone on temperature and its relevance to permeability estimation, *J. Geophys. Res.*, *115*, B09214, doi:10.1029/2010JB007526.

A. Revil, Department of Geophysics, Colorado School of Mines, Green Center, 1500 Illinois St., Golden, CO 80401, USA. (arevil@mines.edu)